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RESEARCH ARTICLE

Explainable spatial machine learning for hedonic real estate modeling

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Abstract

Accurately modeling rents and prices is a key challenge in real estate analysis. Traditional linear models may fail to capture complex non-linear relationships, and spatial dependencies are often ignored in existing machine-learning approaches. This article introduces a novel hybrid statistical machine-learning model for modeling real estate rents and prices. The proposed approach combines a spatial Gaussian process with tree boosting. In so doing, spatial correlations are explicitly accounted for, and the tree-boosting part can handle complex non-linear relationships and interactions. We compare the proposed model against established benchmarks using a large-scale dataset consisting of more than 1.5 million rental apartment listings across Germany and also a smaller condominium price listings dataset. Our findings demonstrate that the proposed model yields superior prediction accuracy due to accounting for both nonlinear patterns and spatial dependencies. We further use machine-learning explainability techniques to better understand the non-linear relationships present among rents and predictor

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variables, and we conduct a detailed analysis of the impact of locational characteristics on rents.

KEYWORDS

Gaussian processes, rental price predictions, spatial dependencies, tree-boosting

1 | INTRODUCTION

The real estate market is a dynamic and complex environment influenced by various factors, including economic conditions, local demand, and neighborhood characteristics. Accurately predicting rental prices is critical for property managers, investors, and policymakers, as it directly impacts housing affordability, investment strategies, and urban development. A reliable assessment of market rents also enables better resource allocation, equitable housing policies, and a more efficient market, benefiting all stakeholders involved. Traditional methods for estimating rents often fall short in capturing complex nonlinear relationships between response and predictor variables, as well as the intricate influence of location-specific attributes.

Advancements in machine learning have markedly enhanced the accuracy of models in across various fields such as finance, healthcare, and real estate. Tree-boosting (Kok et al., 2017; van Wezel et al., 2005), neural networks (Walther & Sigrist, 2019), support vector machines (Lam et al., 2009; Kontrimas & Verikas, 2011), and random forests (Antipov & Pokryshevskaya, 2012; Yoo et al., 2012; Yao et al., 2018) are examples of machine learning models that have been applied in real estate. In general, boosting methods, which iteratively refine predictions by focusing on errors made by previous models, have proven particularly effective in modeling tabular data (Grinsztajn et al., 2022; Januschowski et al., 2022; Nielsen, 2016; Shwartz-Ziv & Armon, 2022). In the context of real estate, recent research by Geerts et al. (2024), which compares various approaches for house price prediction, found that tree-boosting methods consistently outperformed alternative models, including graph neural networks. However, traditional machine-learning methods assume that the response variable is independent conditional on observed covariates. This is an assumption that is likely to often not be fulfilled for real estate rents and prices (Can, 1992; Dubin, 1988).

This article introduces a novel hybrid statistical machine-learning model for modeling real estate rents and prices. Our approach, which is based on Gaussian Process Boosting (GPBoost) (Sigrist, 2022), combines a spatial Gaussian process (GP) (Banerjee et al., 2003; Gelfand et al., 2010) with tree-boosting (Freund et al., 1996; Friedman, 2001). In so doing, the tree-boosting part can handle complex nonlinear relationships and interactions and spatial correlations are explicitly accounted. While gradient-boosted trees are highly effective for general predictive tasks, they can struggle to capture the spatial dependencies inherent in housing data (Hengl et al., 2018; Pace & Hayunga, 2020). By integrating Gaussian processes in tree boosting, GPBoost addresses this limitation, enabling the model to account for intricate spatial patterns while retaining the predictive power of boosting models. A further advantage of the GPBoost approach is that it learns both the Gaussian process hyperparameters and the tree-boosting function jointly without requiring an ad hoc multistep estimation approach. We evaluate the prediction accuracy of the proposed model relative to established benchmark approaches, including traditional linear regression models with and without the incorporation of Gaussian processes and tree boosting. To this end, we use a large-scale dataset of German residential apartment listings with more than 1.5 million samples,



which includes detailed locational and structural characteristics, and a smaller condominium price listing dataset. Moreover, we analyze the model's performance under varying conditions, including settings with limited or no spatial covariates, to assess its robustness across different empirical contexts.

We further use machine-learning explainability techniques to better understand the nonlinear relationships present among rents and predictor variables. Prior studies on rental price modeling in Germany (Bachmann, 2025; Chen et al., 2024; Lorenz et al., 2023) confirm the advantages of machine-learning approaches over traditional linear methods and highlight the growing importance of explainability techniques. Boosting techniques inherently capture nonlinearities and interactions. This is particularly valuable as the individual role and interaction of the over 100 locational characteristics in our rents dataset is *ex ante* not fully understood. Furthermore, the existence of geographic submarkets with location-specific marginal attribute prices (Can, 1992) is addressed as boosting techniques can capture interactions between spatial and structural variables. Model-agnostic techniques such as SHAP (SHapley Additive exPlanations) (Lundberg & Lee, 2017; Lundberg et al., 2020) allow for detailed analysis of feature importance and for showing the individual impact of each variable on the predictions. In addition, SHAP dependence plots can illustrate how the effect of one variable varies across the range of another, providing valuable insights into interactions between variables. This enables not only an understanding of the overall influence of individual variables and broader categories, such as structural, macro-, and micro-location factors, but also an exploration of complex interdependencies, offering a more nuanced explanation of the inner workings of the model and how predictions of rental prices are formed.

Besides Gaussian processes, alternative approaches for modeling spatial dependencies in areal data include conditional autoregressive (CAR) and simultaneous autoregressive (SAR) models (Besag, 1974; Can, 1992; LeSage & Pace, 2009). However, such CAR and SAR models require a user-defined 0–1 or weighted adjacency structure, and they are restricted to areal data. Gaussian processes, on the other hand, can adapt their correlation structure by estimating hyperparameters from data and thus do not require a user to specify a 0–1 or weighted neighborhood structure. Furthermore, Gaussian processes can be used to model both point-referenced and areal spatial data (e.g., by using centroids of areas). For these reasons, we focus on Gaussian processes in this article. The methodology presented in the following could equally well be applied when using, for instance, a CAR model instead of a Gaussian process. Related to our research, recent work of Francke and van de Minne (2024) models spatial dependencies by using an improper intrinsic conditional autoregressive (ICAR) model (Besag & Kooperberg, 1995) and combines it with different machine-learning algorithms using an iterative multistep estimation approach.

The remainder of this article is organized as follows. Section 2 introduces our dataset. In Section 3, we introduce the various model types under consideration, detailing their respective advantages and discussing the unique properties of each model. In Section 4, we present the out-of-sample performance of the different models and key findings, offering a comprehensive analysis of the model behavior using the SHAP framework. We explore the importance of individual variables, examine their impact on predictions, and delve into the interactions between variables to uncover complex patterns. Additionally, we discuss how different models handle these relationships and interactions, providing insights into their relative effectiveness. This section also includes results from various robustness checks to validate the stability and reliability of our findings, including an analysis of robust model versions on a separate dataset of German condominiums offered for sale.

TABLE 1 Overview of data categories.

Categories	Structural	Macro-location	Micro-location	Regionalization
Description	Object-specific variables	Municipality-specific variables	Spatial variables on a 25x25 m raster	Add. municipality-specific variables
Sources	Value AG	FPRE	FPRE	BBSR, FPRE
Nb. attributes	27	35	79	4
Examples	Current net rent	Purchasing power	Road noise night	Kreis type
	Usable space	Property tax rate B	Air dist. to motorway	municipality type
	Number of rooms	Unemployment rate	Slope gradient	
	Renovations	Vacancy rate	Flood risk	
	Year of construction	Single-family home rate	Micro-location rating	
	Floor	Proportion of academics	Subrating Sunlight	

2 | DATA

The analysis is based on listings data of German residential apartments offered for rent. We use data provided by Value AG, which collects real estate listings data from over 100 sources. The examined data range from the second quarter of 2021 to the last quarter of 2022 and contain over 1.5 million listings of rental apartments, which amounts to 6.5% of the rental housing stock.¹ Value AG provides information about the apartment and associated building as well as address information. We geocode the data and add location information provided by Fahrländer Partner Raumentwicklung (FPRE). FPRE obtains these from various public offices and other institutions, prepares them and makes them available in a harmonized and comprehensive manner throughout Germany.

We aim to predict the variable *current net rent* and use a logarithmic transformation for the response variable. Figure 2 illustrates the distribution of the net rent before and after the logarithmic transformation.

For the analysis, we use a total of 149 covariates, which can be grouped in four categories. Table 1 provides an overview of these categories. While our grouping aligns with the common categories defined in hedonic house price models by Dubin (1988) and Can (1992)—structural, location, and neighborhood variables—we refine these definitions to account for differences in spatial granularity and introduce a distinct regionalization category. Specifically, 27 variables contain structural information like the net rent, area, and year of construction.² Thirty-five variables describe the macro-location of the apartment. They include information about purchasing power, taxes, vacancy, population, and employment on municipality-level or higher. Seventy-nine variables include details about the micro-location of the apartment. They are calculated for a raster with a resolution of 25 m and contain information about the amenities in the object's immediate surroundings like the number of grocery stores within a 500 m radius. The individual micro-location indicators were combined to an overall micro-location rating as well as subratings (e.g., for noise or public transport), which are also included in the dataset. We use four variables characterizing the types of municipality or Kreis the apartment is located.³ We call

¹ See Figure B1 in Appendix B for a map of the distribution of rental apartment listings relative to rental housing stock.

² Most of these variables are directly provided by Value AG. Some variables like ZUST, RENOV, and ARTMIETC were constructed as transformations or combinations of the original variables (see Appendix A.1).

³ There are 294 communal associations (Kreise) and 106 independent cities in Germany.

them regionalization variables. Four variables cannot be assigned to any of the aforementioned categories, which are the coordinates, the publication date (year and quarter), as well as dummies to capture regional time trends. In Tables B1 and B2 in the Appendix B, we show a description of all variables with their data types and report the summary statistics of the numeric variables.

We exclude 155 observations because we cannot determine the macro-location (municipality or city district) due to insufficient address information, and we exclude 58 observations because they are located in non-communal areas, where no macro-location information is available. Moreover, 4833 apartments are located outside Germany and are not included in the analysis. Furthermore, 45,410 observations have been flagged by Value AG as either potential input error for the net rent or as price-controlled living spaces (10,058 and 35,352, respectively). Hence, we exclude them from the analysis. All exclusions amount to 3.1% of the original dataset. In total, we use 1,552,649 observations in the analysis.

Due to the nature of listings data, advertisers can choose which information they want to publish. Therefore, we face missing values in various attributes, reflecting a general challenge that arises in almost every field of applied statistics (Josse & Reiter, 2018). Missing values in categorical variables are treated as a new category. All other variables are imputed using the iterative imputation method called *missForest* introduced by Stekhoven and Bühlmann (2012), which is based on a random forest. The algorithm can handle mixed-type data and delivers particularly good results relative to other imputation methods when complex relationships and nonlinearities are present. Due to computational constraints, we limit the number of trees grown in each forest to 10. As demonstrated by Stekhoven (2011), this restriction can still yield reliable and appropriate results. Additionally, the imputation process is carefully integrated within the train-test split to avoid data leakage and ensure accurate evaluation. Table B2 in Appendix B shows the number of observations per variable that have been imputed using the *missForest* algorithm.⁴ While simply discarding observations with missing values might appear tempting, this can introduce biases due to potential systematic missingness patterns. However, in Section 4.5.1, we present results based on the non-imputed data by discarding all observations with missing values, in order to assess the robustness of our findings to the imputation procedure.

Besides having missing values in the above-mentioned covariates, we can determine the exact location only for 48% and an approximate location (e.g., a street name) for 16% of the observations. We use the coordinates of the center of the municipality or city district for the remaining observations.⁵ As the exact location is only available for around half of the data, we present the following results for both the dataset with all observations and the one with only exact locations. However, using only the reduced dataset affects the spatial coverage of the domain (Germany). In Figure 1, we show that the exact address is provided more often in eastern Germany and in urban areas. Nevertheless, the subset analysis can deliver additional insights as we plan to investigate the importance of micro-location specific attributes that have to be imputed in case of insufficient address information. Doing the analysis on both datasets, we can further evaluate whether these findings are robust to the imputation procedure.

⁴ Year of construction is an important variable for accurately modeling residential real estate data. We have to impute it for 44% of the observations. For further information on the imputation and its consequences, see Appendix A.2.

⁵ We matched the micro-location data only for observations with at least an approximate location and treated the unmatched data as missing values to impute in the next step.

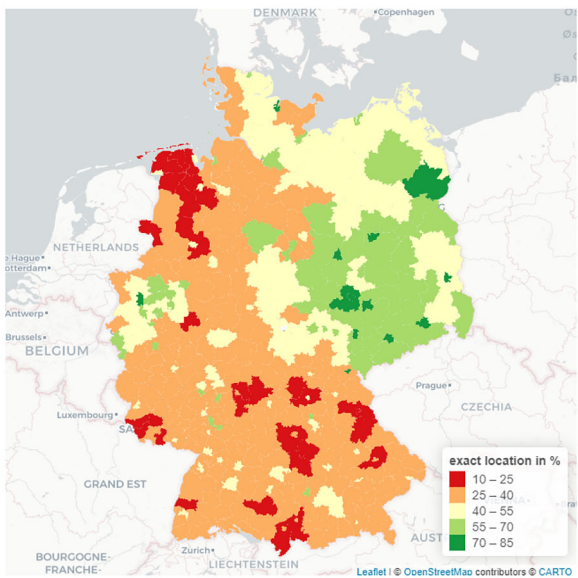


FIGURE 1 Share of available exact location on Kreis level.

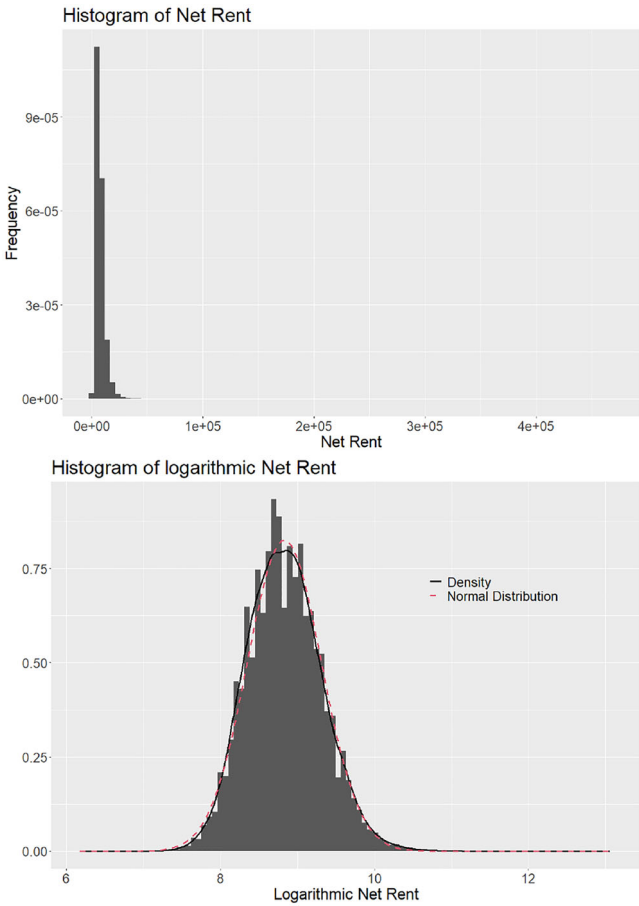


FIGURE 2 Distribution of the raw net rent and the logarithmic net rent.

3 | MODELS

This section presents the various model types considered in this study. We assume the following general mixed effects model

$$\mathbf{Y} = F(\mathbf{X}) + \mathbf{b} + \epsilon, \quad \mathbf{b} \sim \mathcal{N}(\mathbf{0}, \Sigma), \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n),$$

where $\mathbf{Y} = (Y(\mathbf{s}_1), \dots, Y(\mathbf{s}_n))^T \in \mathbb{R}^n$ represents the response variable, which in our case is the logarithmic net rent observed at locations $S = \{\mathbf{s}_1, \dots, \mathbf{s}_n\}$, $\mathbf{s}_i \in \mathbb{R}^2$, $F(\mathbf{X}) \in \mathbb{R}^n$ denotes the fixed effects, and the design matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ contains all the variables described in the previous section for the n observations. Additionally, the error term $\epsilon = (\epsilon(\mathbf{s}_1), \dots, \epsilon(\mathbf{s}_n))^T \in \mathbb{R}^n$ represents independent noise, commonly referred to as the nugget effect. The zero-mean random effects $\mathbf{b} = (b(\mathbf{s}_1), \dots, b(\mathbf{s}_n))^T \in \mathbb{R}^n$ are modeled as a finite-dimensional realization of a GP, with a corresponding covariance matrix $\Sigma \in \mathbb{R}^{n \times n}$.

Incorporating a GP into the model allows for modeling spatial dependencies, which enables smooth, continuous spatial predictions and reduces bias in the estimates of the parameters characterizing the fixed effects $F(\cdot)$. Additionally, this integration allows for probabilistic predictions, providing an uncertainty measure for each forecast and offering a more comprehensive assessment of prediction reliability (Cressie, 1993; Rasmussen & Williams, 2006). However, a limitation of Gaussian processes is their scalability to large datasets. To address this issue, we use an approach introduced in Gyger et al. (2024) that applies iterative methods in a full-scale GP approximation to accelerate computations. This approximation includes a low-rank predictive process component that captures large-scale spatial dependencies and a compactly supported residual component that accounts for remaining short-range dependencies (Sang & Huang, 2012). The approximation is governed by two key parameters: the number of inducing points $m \in \mathbb{R}$, which determines the rank of the predictive process part, and the taper range $\gamma \in \mathbb{R}$, which defines the maximum distance beyond which the residual correlation is set to zero. In our models, we use $m = 500$ inducing points, selected using the k-means++ algorithm (Arthur & Vassilvitskii, 2007), and a taper range that results in an average of approximately 80 non-zero entries per row in the residual covariance matrix. This setup ensures that our models effectively capture both large-scale and short-scale spatial dependencies.

3.1 | Linear Gaussian process model

Linear models

$$\mathbf{Y} = \bar{\mathbf{X}}\boldsymbol{\beta} + \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n). \quad (1)$$

are the most widely used type of hedonic real estate models. These models assume linear fixed effects of the form $F(\mathbf{X}) = \bar{\mathbf{X}}\boldsymbol{\beta}$, where $\boldsymbol{\beta} \in \mathbb{R}^{p+1}$ is the vector of coefficients, and $\bar{\mathbf{X}} \in \mathbb{R}^{n \times (p+1)}$ represents the design matrix $\mathbf{X}$ augmented with an intercept term to account for the baseline level of the outcome.

As previously mentioned, we also include a random effect term, modeled as a finite-dimensional GP:

$$\mathbf{Y} = \bar{\mathbf{X}}\boldsymbol{\beta} + \mathbf{b} + \epsilon, \quad \mathbf{b} \sim \mathcal{N}(\mathbf{0}, \Sigma), \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n). \quad (2)$$

To mitigate the constraints of the linearity assumption, we include common transformations of the predictor variables. Specifically, we applied the logarithm to all area and distance variables. For variables associated with duration and those representing numbers per unit or area, we used the square root transformation. Additionally, we apply a composition of the arcsine and square-root functions to proportions. Lastly, we add the squared year of construction as a new variable to capture potential nonlinear effects related to the age of the construction as implemented by Case et al. (2004) and Fahrlander (2006). For a detailed overview, we refer to Table B4 in Appendix B.

3.2 | Linear fixed effects model

For comparison with the GP in modeling spatial dependencies, we incorporate fixed effects location dummy variables for the spatial units, denoted as $\alpha = (\alpha_1, \dots, \alpha_n)^T \in \mathbb{R}^n$, into the linear model to capture unobserved location-specific factors (Allison, 2009):

$$Y = \bar{X}\beta + \alpha + \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 I_n). \quad (3)$$

This spatial unit α typically represents a village, hamlet, or city quarter and divides Germany into 13,058 connected areas.

3.3 | Gradient tree boosting

Boosted regression trees (Breiman, 1998; Bühlmann & Hothorn, 2007; Freund et al., 1996; Friedman, 2001; Sigrist, 2021) are widely recognized for their exceptional predictive accuracy with tabular data (Grinsztajn et al., 2022; Januschowski et al., 2022; Nielsen, 2016; Schwartz-Ziv & Armon, 2022). Boosting combines multiple base learners in an additive manner, resulting in

$$F(\mathbf{X}) = \nu \cdot \sum_{k=1}^K \mathcal{T}_k(\mathbf{X}), \quad (4)$$

where $\mathcal{T}_k$ represents a base learner, and ν denotes the associated weight for the model, also known as the learning rate. Base learners can be any predictive model, but regression trees are used as the base learners in the case of gradient-boosted trees. The function $F(\cdot)$, which maps the p predictors to a scalar prediction, is determined by minimizing an appropriate loss function such as the L2 loss, which is used in this article. For this loss function, the k -th base learner is the least squares approximation to the residuals from a model with $k - 1$ base learners. Since the gradient of the L2 loss function is equivalent to the residuals, this process can be interpreted as gradient descent in a functional space.

Boosted trees offer several advantages. They excel at capturing complex interactions and nonlinear relationships within the data. Additionally, they are inherently resilient to multicollinearity, ensuring that correlated predictor variables do not adversely affect model performance. Furthermore, boosted trees are invariant to monotone transformations of predictor variables, which enhances their robustness and flexibility in modeling. Unlike linear models, this property eliminates the need for pre-transformations of predictor variables.

Tree boosting has a few tuning parameters. The tuning parameters that characterize the trees include the maximum depth of each tree, which regulates its complexity, and the minimum number of data points per leaf, which ensures each leaf has enough data. Additionally, we consider the

TABLE 2 Tuning parameter grid for boosted trees.

Hyperparameter	Values
Maximal depth	5, 10
Minimal data in leaf	5, 10
Learning rate	0.1, 0.01
L2 regularization	1, 10

learning rate ν , L2 regularization of the leaf coefficients, and the number of boosting iterations K . For the number of boosting iterations, an early stopping mechanism is implemented: The number of iterations K is capped at 1000, and the fitting process is halted if the validation loss does not improve for 10 consecutive iterations. Unless stated otherwise, the tuning parameter grid in Table 2 was used to find the optimal configuration minimizing the MSE on a validation set.

Additionally, we can incorporate monotone constraints into tree boosting to enforce a prior belief that the true relationship between specific predictors and the response is either monotone increasing or decreasing. This method helps the model better reflect anticipated relationships and lowers the chance of unexpected prediction outcomes.

3.4 | Combining gradient tree boosting with gaussian process models

The GPBoost algorithm introduced in Sigrist (2022) combines tree boosting with GP models to leverage the strengths of both methods and address their individual limitations

$$\mathbf{Y} = \nu \cdot \sum_{k=1}^K \mathcal{T}_k(\mathbf{X}) + \mathbf{b} + \epsilon, \quad \mathbf{b} \sim \mathcal{N}(\mathbf{0}, \Sigma), \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n). \quad (5)$$

The GPBoost algorithm iteratively learns the covariance parameters of the GP while adding trees to the ensemble using gradient boosting steps. For selection tree-boosting related tuning parameters, we used the parameter grid in Table 2.

3.5 | Training, validation, and test data

To evaluate and compare the predictive performance of the models, we use the first six quarters (Q2 2021 to Q3 2022) as the training data and reserve the final quarter (Q4 2022) as the test data. We then compute the root mean squared error (RMSE) in an out-of-sample manner by comparing the predicted logarithmic net rent to the actual logarithmic net rent

$$\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2},$$

where $y_i = \log(\text{rent}_i)$ and $\hat{y}_i = \widehat{\log(\text{rent}_i)}$. For evaluating predictive distributions, we additionally calculate the continuous ranked probability score (CRPS) defined as

$$\frac{1}{n_p} \sum_{i=1}^{n_p} \bar{\sigma}_i \left(\frac{-1}{\sqrt{\pi}} + 2 \cdot \Phi \left(\frac{y_i - \hat{y}_i}{\bar{\sigma}_i}; 0, 1 \right) + \frac{y_i - \hat{y}_i}{\bar{\sigma}_i} \left(2 \cdot \Phi \left(\frac{y_i - \hat{y}_i}{\bar{\sigma}_i}; 0, 1 \right) - 1 \right) \right),$$

where $\phi(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ is the density function and $\Phi(x; \mu, \sigma)$ is the cumulative distribution function of a normal distribution $\mathcal{N}(\mu, \sigma)$, n_p is the number of prediction locations and $\hat{\sigma}$ is the predictive variance. Comparing both RMSE and the CRPS offers a more comprehensive evaluation of model performance. While RMSE provides insights into the accuracy of point predictions, the CRPS assesses the quality of probabilistic forecasts.

To select the tuning parameters of the machine learning models, we use cross-validation, setting aside the last quarter of the training data (Q3 2022) as the validation set, corresponding to the leave-time-out cross-validation approach discussed in Schratz et al. (2024). The model is tuned on the remaining training data (Q2 2021 to Q2 2022) by maximizing predictive accuracy on the validation data. Once all tuning parameters are chosen, the machine learning models are trained on the full training data (Q2 2021 to Q3 2022). This validation design is consistent with the discussion in Deppner and Cajias (2024), which emphasizes that the choice of cross-validation scheme should reflect the intended prediction goal.

For the robustness checks in Section 4.5, we adopt an expanding window approach. In each iteration, we extend the training window by incorporating an additional quarter of the data while keeping the validation set fixed as the last quarter of the training data. The test data always consist of the quarter immediately following the training window. Additionally, we consider a random train-test split, selecting 80% of the entire dataset as the training data and the remaining 20% as the test data. The same procedure is applied for tuning the machine learning models, where 20% of the training set is randomly selected as the validation set.

4 | RESULTS

In this section, we present the results of our analysis, starting with a comparison of linear models with and without Gaussian processes in predicting net rental prices. The aim is to explore the role of Gaussian processes in capturing spatial dependencies that are important in real estate data. We then provide the main comparison, evaluating the predictive performance of a linear GP model (2), an independent boosting model (4), and a GPBoost model (5) across different datasets to explore the strengths and limitations of each approach. We utilize both precisely geocoded data and all geocoded data to ensure comprehensive spatial coverage across Germany. Additionally, we train and test the models with and without spatial variables—such as micro-location, macro-location, regionalization factors, and regional time trends—to evaluate their performance in the absence of these variables and to minimize dependence on imputed data in the all geocoded scenario. Furthermore, we assess the impact of incorporating monotone constraints in boosting models and their effect on accuracy. Finally, we conduct robustness checks for different train-test splits to further validate model performance.

4.1 | Modeling spatial dependencies

We analyze the effectiveness of Gaussian processes in capturing spatial variation by comparing the predictive performance when predicting rental prices of linear models with and without the inclusion of a GP (linear GP model (2) and linear model (1)). To further assess the importance of spatial modeling in real estate, we compare the results to a model that includes spatial units as fixed effects (linear FE model (3)). We cannot use the linear FE model (3) to predict spatial

TABLE 3 Root mean square error (RMSE) comparison across linear models and data versions (percentage decrease in parentheses).

Data	Linear model	Linear GP model	Linear FE model
Exact geocoding	0.17301 (Benchmark)	0.15817 (−8.58%)	0.16694 (−3.51%)
Exact geocoding without spatial variables	0.28088 (Benchmark)	0.16152 (−42.50%)	0.16936 (−39.70%)
All geocoding	0.18403 (Benchmark)	0.17400 (−5.45%)	0.17736 (−3.62%)
All geocoding without spatial variables	0.28331 (Benchmark)	0.17660 (−37.67%)	0.17886 (−36.87%)

Note: Without spatial variables means that we omitted all variables related to micro-location, macro-location, regional distinctions, and regional time trends (AQREG_KR).
Abbreviation: GP, Gaussian process.

units that were not included in the training set, so we limit the test set for all models. Specifically, we remove 0.10% of the data points from the general test set and 0.23% from the exact geocoding test set.

The results in Table 3 show that the linear GP model (2) is more accurate. For the full dataset, it reduces the RMSE of the linear model (1) by 5.45% and by 37.67% when spatial variables are excluded. This suggests that the GP significantly enhances the model’s ability to capture spatial dependencies. While the linear FE model (3) also demonstrates improvement over the pure linear model (1), with RMSE reductions of 3.62% for the complete dataset and 36.87% when spatial variables are excluded, it remains less accurate than the linear GP model (2).

Furthermore, for the linear FE model (3), we train the model by excluding all training points associated with spatial units that occur fewer than three times (0.15% for general geocoding and 0.43% for exact geocoding) due to insufficient data to ensure reliable modeling. The prediction error remains essentially unchanged (the results are not tabulated).

To assess whether the observed differences in predictive accuracy are statistically significant, we employ a mixed-effects model. Specifically, we estimate a model where the squared residuals serve as the response variable, including fixed effects for model type and a random effect for each observation to account for the potential correlation in predictions for the same rental apartment across models. Model type is dummy-encoded, with the linear GP model (2) as the reference category. The results of the mixed-effects model indicate that all differences in predictive accuracy between models are statistically significant. The full statistical results, including the significance levels and detailed estimates, are provided in Appendix B Table B6. These results show that the improvements of the linear GP model (2) over both the linear model (1) and the linear FE model (3) are statistically significant at the 1% level across all data versions.

4.2 | Main results

Table 4 shows the RMSE and CRPS of different models for different datasets and predictor variable sets. For each dataset, the boosting model (4) and GPBoost model (5) show improvements over the baseline model, linear GP model (2), with the percentage decrease in RMSE highlighted in parentheses.

The GPBoost model (5) consistently outperforms other models, exhibiting the lowest RMSE and CRPS across various data configurations. Specifically, with exact geocoding, the boosting model (4) reduces the RMSE and CRPS of the linear GP model (2) by 5.76% and 3.19%, respectively, while the GPBoost model (5) achieves reductions of 6.42% and 6.11%. Excluding spatial variables, the

TABLE 4 Root mean square error (RMSE) and continuous ranked probability score (CRPS) of various models and data versions (percentage decrease of RMSE and CRPS in parentheses).

Data	Performance measure	Linear GP model	Boosting model	GPBoost model
Exact geocoding	RMSE	0.15828 (Benchmark)	0.14916 (−5.76%)	0.14812 (−6.42%)
	CRPS	0.08577 (Benchmark)	0.08303 (−3.19%)	0.08053 (−6.11%)
Exact geocoding without spatial variables	RMSE	0.16168 (Benchmark)	0.16002 (−1.03%)	0.15601 (−3.51%)
	CRPS	0.08781 (Benchmark)	0.08759 (−0.25%)	0.08503 (−3.17%)
All geocoding	RMSE	0.17405 (Benchmark)	0.16823 (−3.34%)	0.16817 (−3.38%)
	CRPS	0.09507 (Benchmark)	0.09198 (−3.25%)	0.09188 (−3.36%)
All geocoding without spatial variables	RMSE	0.17667 (Benchmark)	0.16953 (−4.04%)	0.16747 (−5.21%)
	CRPS	0.09665 (Benchmark)	0.09274 (−4.05%)	0.09122 (−5.62%)

Note: Without spatial variables means that we omitted all variables related to micro-location, macro-location, regional distinctions, and regional time trends (AQREG_KR).

Abbreviation: GP, Gaussian process.

boosting model (4) shows marginal improvement over the linear GP model (2), with reductions in RMSE and CRPS of only 1.03% and 0.25%, indicating its limited ability to capture spatial dependencies. In contrast, the GPBoost model (5) achieves significant reductions of 3.51% and 3.17% in RMSE and CRPS, respectively, as the GP component effectively captures spatial variations. With all geocoding data, the boosting model (4) achieves reductions of 3.34% in RMSE and 3.25% in CRPS compared to the linear GP model (2), while the GPBoost model (5) achieves similar results with reductions of 3.38% and 3.36%, respectively. Notably, when spatial variables are excluded, the GPBoost model (5) outperforms the boosting model (4) by reducing RMSE and CRPS by 5.21% and 5.62%, compared to 4.04% and 4.05%, respectively.

The impact of imputed data on the results is evident in the comparison between models with and without spatial variables for all geocoding. For the boosting model (4), the difference in performance between including and excluding spatial variables is minimal, suggesting that the spatial information, much of which is imputed, provides little added value. Interestingly, for the GPBoost model (5), excluding the spatial variables actually results in better performance. This indicates that the imputed spatial data may introduce noise, hindering the model's ability to leverage the spatial relationships effectively through the variables.

To evaluate if the differences in predictive accuracy are statistically significant, we utilize a mixed-effects model with squared residuals as the response variable, as described in Section 4.1. Fixed effects for model type (dummy-encoded, with the boosting model (4) as the reference category) and random effects for observations account for correlation across models. The fixed-effects coefficients, t-values, and p-values are shown in Table 5. The results indicate that the linear GP model (2) consistently underperforms compared to the boosting model (4) across all data versions. Furthermore, the improvement of the GPBoost model (5) over the boosting model (4) is

TABLE 5 Test results for comparing the different models. The tests are done using a regression model with a random effect at the observation level.

Data	Summary	(Intercept)	Linear GP model	GPBoost model
Exact geocoding	Estimate	0.02225	0.00280	−0.00031
	<i>t</i> value	134.87	32.27	−3.55
	<i>p</i> value	0.00***	8.74e −228***	3.87e −04***
Exact geocoding without spatial variables	Estimate	0.02561	0.00053	−0.00127
	<i>t</i> value	146.77	4.86	−11.55
	<i>p</i> value	0.00***	1.19e −06***	7.47e −31***
All geocoding	Estimate	0.02830	0.00199	−0.00002
	<i>t</i> value	213.65	34.01	−0.36
	<i>p</i> value	0.00***	3.69e −253***	7.17e −01
All geocoding without spatial variables	Estimate	0.02874	0.00247	−0.00069
	<i>t</i> value	215.99	41.16	−11.52
	<i>p</i> value	0.00***	0.00***	1.09e −30***

Note: Without spatial variables means that we omitted all variables related to micro-location, macro-location, regional distinctions, and regional time trends (AQREG_KR).
Abbreviation: GP, Gaussian process.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

statistically significant at the 1% level, except when using the full dataset. This may be because tree boosting effectively captures most of the spatial variation through the spatial variables, leaving little room for the GP to model additional spatial dependencies, especially when many data points share the same location.

Previous studies have demonstrated that gradient boosting outperforms commonly used methods, such as linear regression, generalized additive models, spatial autoregressive regression, Gaussian regression, and graph neural networks, in predicting property prices and rents (Geerts et al., 2024; Lorenz et al., 2023). However, to the best of our knowledge, we are the first to apply GPBoost in this context and show that it further improves upon gradient boosting. This enhancement can be attributed to the fact that tree-based models leave spatial dependencies in the residuals unutilized for prediction (Hengli et al., 2018; Pace & Hayunga, 2020).

4.2.1 | Economic relevance and prediction accuracy

To further evaluate the models' performance, we will also include an additional metric, the Mean Absolute Percentage Error (MAPE)

$$\text{MAPE} = 100 \cdot \frac{1}{n} \sum_{i=1}^n \left| \frac{\text{rent}_i - \widehat{\text{rent}}_i}{\text{rent}_i} \right|,$$

where $\widehat{\text{rent}}_i = \exp \left(\hat{y}_i + \frac{\hat{\sigma}_i}{2} \right)$. The MAPE is particularly important for assessing the economic relevance of the model errors, as they provide an intuitive measure in percentage terms. We note that the boosting models (4) and (5) were trained and tuning parameters were chosen using the MSE instead of the MAPE, which emphasizes minimizing larger errors. As a result, the models

TABLE 6 Comparison of model performance (exact geocoding).

Metric	Linear model	Linear FE model	Linear GP model	Boosting model	GPBoost model
MAPE	13.31%	12.71%	11.74%	10.98%	10.86%
MAPE applied to avg. monthly rent in EUR	86.33	82.44	76.14	71.22	70.44
MdAPE	10.39%	9.86%	8.81%	8.29%	8.11%
Within 20%	78.7%	80.5%	83.3%	85.3%	85.7%
R^2	0.877	0.886	0.897	0.909	0.910

Note: All measures were calculated on the data with exact geocoding, including spatial variables. The measures for other data versions can be found in Appendix B Table B7.

Abbreviations: GP, Gaussian process; MAPE, mean absolute percentage error; MdAPE, median absolute percentage error.

focus on reducing significant deviations, potentially at the expense of accuracy for smaller values. As noted in Section 4.1, the linear FE model (3) cannot predict outcomes for spatial units absent from the training set, so we adjust the test set accordingly for all models.

Table 6 shows various performance metrics for our models. These metrics are used in similar studies of hedonic models such as Lorenz et al. (2023), Bastian Krämer and Schäfers (2023), and Stang et al. (2023). The GPBoost model (5) outperforms all other models regardless of the choice of performance metric. Its MAPE amounts to 10.86%, which is around 1 pp lower than for the linear GP model (2) and almost 2.5 pp lower than for the linear model (1). Applied to the average monthly rent in the dataset of 648.49 EUR, the error is reduced by approximately 16 EUR from 86 to 70 EUR when using the GPBoost model (5) instead of the linear model (1). The median absolute percentage errors (MdAPEs) show a pattern similar to the MAPE but at a lower level. 85.7% of the predictions in the test sample lie within $\pm 20\%$ of the observed rent. The performance of the boosting models 4 and 5 is at least as good than the results of other gradient boosting models that have been applied to the data of German rental or residential properties (Bastian Krämer & Schäfers, 2023; Lorenz et al., 2023; Stang et al., 2023). The improvement in MAPE of the GPBoost model (5) relative to the boosting model (4) amounts only to 0.12 pp and represents the smallest difference between any two models. However, small but persistent increases in prediction accuracy can still be relevant given the 23 million rental apartments in Germany that could be valued by these models.

4.3 | Model explainability

Our results suggest that the models using gradient tree boosting surpasses the linear models in terms of prediction accuracy. However, linear models can be easily interpreted by looking at parameter estimates. To interpret the predictions of the boosting models, we use a framework based on cooperative game theory called SHAP (SHapely Additive exPlanations) introduced by Lundberg and Lee (2017) and further developed by Lundberg et al. (2020). We analyze variable importance, partial dependencies, and interaction effects using this framework. The SHAP values can be calculated for every observation and describe the contribution of a single variable to the difference between the individual prediction and the expected prediction for the entire dataset. We assess the variable importance by using the mean absolute SHAP value, which indicates the average contribution of a feature to the predictions.

Figure 3 shows the average SHAP values. We find that *usable space* is by far the most important variable. It affects the prediction on average by 25% in both models. Other structural variables such as *heating system*, *standard*, *condition*, and *year of construction* are among the most



FIGURE 3 SHAP (SHapley Additive exPlanations) variable importance (all geocoding). The numbers in parentheses indicate data categories: (1) structural data, (2) macro-location data, (3) micro-location data, (4) regionalization, and (5) other.

important features. However, their average impact on prediction is much smaller. In the boosting model (4), macro-location variables such as *annual wage/income tax per taxpayer*, *vacancy rate*, *purchasing power per inhabitant*, and *proportion of academics* contribute strongly to the prediction. These spatial variables are much less important in the GPBoost model (5). We interpret this in the sense that the boosting model uses the observable macro-location variables to compensate

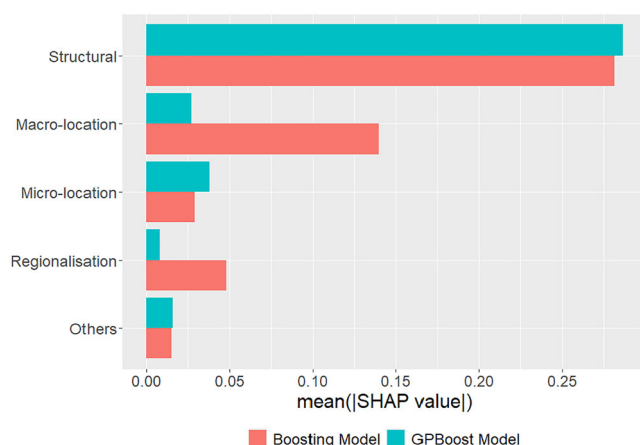


FIGURE 4 SHAP (SHapley Additive exPlanations) variable importance per category (all geocoding).

for spatial effects, which are modeled with the GP in the GPBoost model. Note that the random GP models variability that is not explained by the fixed effects variables. Consequently, if the macro-location variables were truly related to the response variable to such a degree as in the boosting model, they would also be more heavily in the GPBoost model. This finding is also in line with the above results that the GPBoost model has higher prediction accuracy compared to the boosting model. In the boosting model (4), neither individual micro-location variables nor the coordinates contribute much to the prediction. However, the coordinates become the most important variables after *usable space*, once the other spatial variables are removed (results not shown). Looking at the dataset using only exact locations, we see almost the same patterns among the most important variables as when looking at all data (see Figure B2 in Appendix B).

Figure 4 shows the cumulated SHAP variable importance for each covariate category. Structural variables are the most important covariates. We observe that macro-location data are second most important in the independent boosting model. Furthermore, although individual micro-location variables appear rarely among the top 30 most important variables, we observe that the 79 variables together can have a noticeable contribution. For instance, in the GPBoost model, they are the second most important category of covariates. In the plain boosting model, regionalization covariates play an important role, particularly because the variable *Federal state number with municipality type* ranks third in importance. However, its contribution is substantially reduced when the GP component is added in the GPBoost model.

In the linear GP model (2), complex nonlinear relationships and interactions between predictors are not accounted for. However, SHAP interaction effects and dependence plots derived from the boosting model (4) clearly reveal the presence of such patterns in the data. This suggests that the linear GP model may fail to fully capture the relationships between variables. To address this limitation and to better understand the role of spatial information, we perform an additional analysis to quantify the impact of both micro-location and macro-location data on prediction accuracy. Table 7 shows the change in prediction accuracy (RMSE) when removing spatial variables from the data with exact geocoding. These results, partially derived from Table 4, reveal that while the increase in RMSE for the linear GP model (2) is relatively modest at only 2.1%, the spatial variables have a significantly stronger impact on the prediction accuracy in the boosting model (4), leading to an increase of 7.3%. Approximately half of this effect can be attributed to micro-location variables (+3.7%). Interactions and non-linearities among the spatial variables can be captured by the boosting model (4) increasing the value of having them in the model.

TABLE 7 Change in the root mean squared error (RMSE) when removing spatial variables.

Data	Linear GP model	Boosting model
Exact geocoding without all spatial variables	2.1%	7.3%
Exact geocoding without micro-location var.	1.0%	3.7%

Note: Removing spatial variables means that we omitted all variables related to micro-location, macro-location, regional distinctions, and regional time trends (AQREG_KR).
Abbreviation: GP, Gaussian process.

TABLE 8 Change in root mean squared error (RMSE) when using GPBoost instead of the linear Gaussian process (GP) model for various apartment segments.

Segment	RMSE improvement
Baseline	3.4%
Year of construction > 2012	10.3%
Year of construction < 1934	3.6%
Usable space > 92	6.0%
Usable space < 34	8.6%

Note: We use the data version with all geocoding and all spatial variables.

The SHAP dependence plots in Figure 5 show four important structural variables plotted against their respective SHAP values. On average, the impact of these variables on the predictions is as expected. Regarding the transformations used in the linear GP model (2), we see that using a logarithmic transformation for *usable space*, separate factor levels for *standard* and *condition*, and introducing a quadratic term for *year of construction* were sensible choices. Taking a closer look at the plot for *usable space*, we see that for the smallest and largest apartments, the logarithmic transformation is not sufficient to fully capture the relationship. Fahrländer (2006) supports this finding when analyzing Swiss residential properties. He shows that prediction accuracy can be improved when using a piecewise log-linear term for the area of the apartment. Modeling *year of construction*, the quadratic fit also does not seem to capture all of the nonlinear relationship. Similar to Stevenson (2004), who uses a cubed term of age, including an additional cubed term of *year of construction* in a linear model clearly improves the fit. As boosting models are superior in capturing these complex nonlinear relationships, they improve prediction accuracy, especially in the apartment segments that show high deviations between the LOESS line (in black) and the fit from a linear model (in red). We illustrate this further in Table 8 that shows that the improvement in RMSE of (5) over (2) is higher in these apartment segments.

Furthermore, SHAP plots are a useful tool for uncovering heterogeneity in local effects that can arise due to interactions. This is visible by looking at the spread of SHAP values along the y-axis for a given feature value. As the scales for the y-axis in Figure 5 differ for the four subplots, one might think at first sight that the heterogeneity for *usable space* is negligible. However, the biggest interaction effects can actually be found for *usable space*, and they are even higher than the main effect of the other three displayed structural variables.

Figure 6 shows the SHAP dependence of one of the most important macro-location variable *annual wage/income tax per taxpayer*. The dispersion of points along the y-axis is quite high, especially with respect to the mean absolute SHAP value of 0.046. This suggests that the interaction effects are high relative to the average contribution of this variable. The coloring of the figure reveals an interaction with *usable space*, suggesting that taxes have a larger impact on the

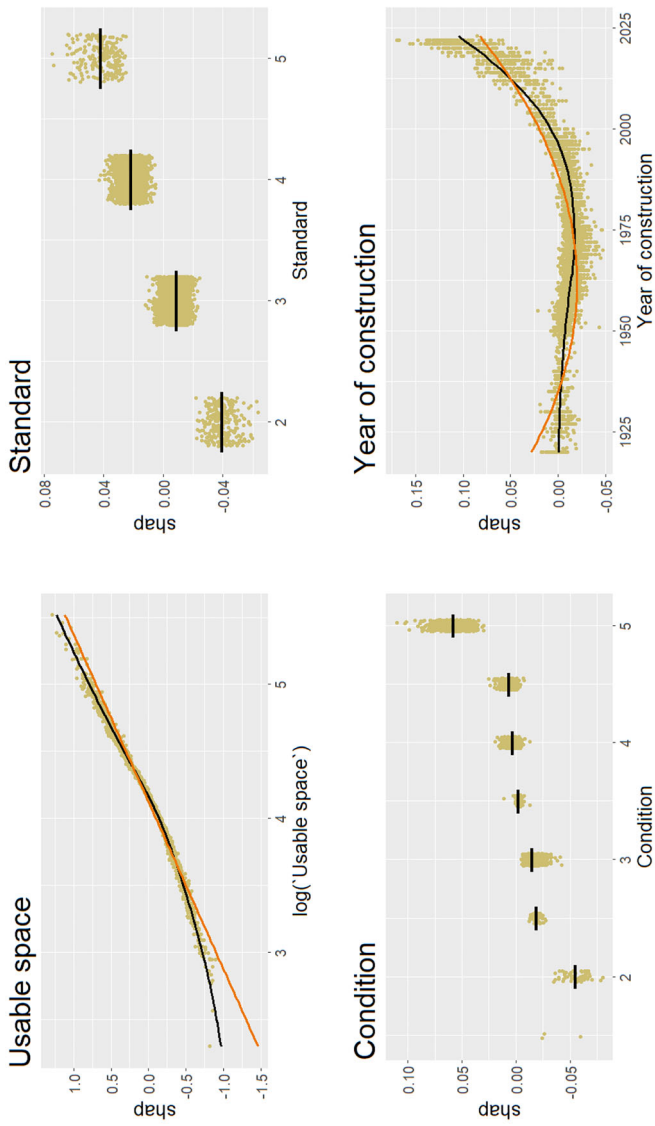


FIGURE 5 SHAP (SHapley Additive exPlanations) dependence plots for important structural data. The plots show SHAP values generated for the GPBoost model (5) and were calculated on a random subsample of 5000 observations to reduce computing time. For “usable space” and “year of construction,” a LOESS (locally estimated scatterplot smoothing) line (in black) highlights the underlying trends. Furthermore, the orange lines depict a linear fit for “usable space” and a quadratic fit for “year of construction.” Additionally, for the categorical variables “standard” and “condition,” the group-wise means (in black) are illustrated.

FIGURE 6 Interactions: SHAP (SHapley Additive exPlanations) dependence for taxes colored by usable space. The plots show SHAP values generated for the boosting model (4) and were calculated on a random subsample of 5000 observations to reduce computing time.

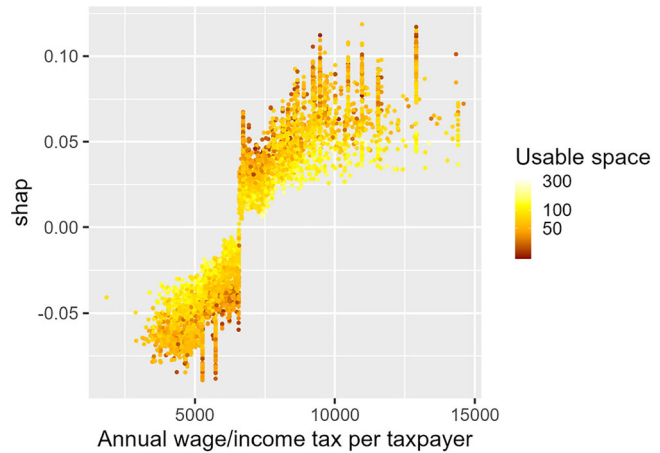
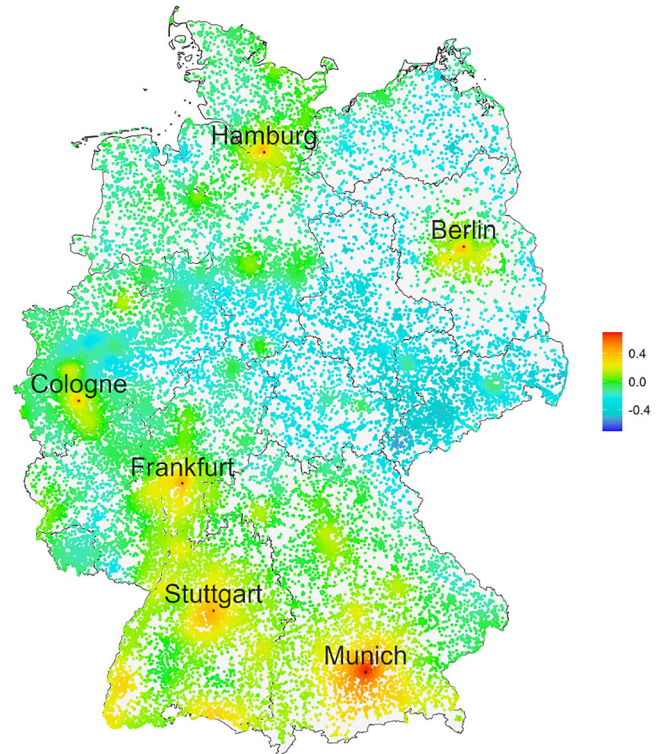


FIGURE 7 Estimated Gaussian process (GP) random effects of the GPBoost model (5).



prediction for small than for large apartments. Furthermore, the tax variable also features relatively strong interactions with *proportion of academics* and *federal state number with municipality type*. Generally, most important macro-location variables feature relatively strong interaction effects with either *usable space* or other macro-location variables, as indicated by the SHAP interaction values (results not shown).

In addition to the explainability of the boosting component using SHAP values, the GP part of the GPBoost model provides insight into the underlying spatial structure of the data. Figure 7 visualizes the mean of the estimated GP random effects from the GPBoost model, highlighting its role in capturing residual spatial dependence. In addition, Table B5 in Appendix B reports the

TABLE 9 Root mean squared error (RMSE) and continuous ranked probability score (CRPS) for data with exact geocoding (percentage increase of RMSE and CRPS in parentheses).

	Performance measure	Boosting model	Boosting with constraint version 1	Boosting with constraint version 2
Test	RMSE	0.14916 (Benchmark)	0.15060 (+0.97%)	0.15182 (+1.78%)
	CRPS	0.08303 (Benchmark)	0.08375 (+0.87%)	0.08342 (+0.47%)
Train	RMSE	0.08927 (Benchmark)	0.09128 (+2.25%)	0.10546 (+18.14%)

estimated covariance parameters for both the linear GP and GPBoost models, quantifying the strength and range of the spatial dependence.

4.4 | Expert constraints

By incorporating expert knowledge into tree-boosting models through monotone constraints, we can enforce prior beliefs about the relationship between specific predictors and the target variable. This ensures that these relationships follow a monotonic trend, either increasing or decreasing. Constrained models could offer advantages, especially when explainability or adherence to domain-specific knowledge is important. By enforcing monotonicity, the model aligns with prior expectations about how certain features should influence the target variable, which can improve trust and transparency in predictions, even if it sacrifices a small amount of accuracy. We explore two variations of expert constraints: one focusing solely on a few structural variables and the micro-location rating, and another one that also incorporates some micro-location variables. For a detailed breakdown of these variables and their respective constraints, we refer to Table B3 in the Appendix B. In this comparison, we focus on the independent boosting approach utilizing data with exact geocoding. While monotonic constraints on the fixed effects are straightforward to apply and supported in the GPBoost implementation, we exclude the GPBoost model from this analysis due to runtime considerations and simplicity.

Table 9 shows that the boosting models with monotone constraints perform slightly worse than the unconstrained model in terms of RMSE. The observed differences in RMSE are statistically significant at the 1% level, as detailed in Table B8 in Appendix B. However, despite not achieving better predictive accuracy, the performance gap between the models is relatively small. Notably, the training RMSE differences between the constrained and unconstrained models are substantially larger than the test RMSE differences. This phenomenon can be explained by the spiked-smooth interpolation properties of tree-ensembles, as discussed in Wyner et al. (2017). Tree-based methods tend to interpolate the data by closely fitting certain *spiked* regions, such as outliers, while maintaining a smoother fit elsewhere. In the constrained models, the monotonicity restrictions limit the model's flexibility to interpolate these spiked regions during training, leading to higher training RMSEs. However, the smoother generalization achieved through these constraints helps the models perform comparably well on the test data, reducing the RMSE differences out-of-sample.

TABLE 10 Root mean squared error (RMSE) and continuous ranked probability score (CRPS) of various models and for the non-imputed dataset, where all rows containing missing values were deleted (percentage decrease of RMSE and CRPS in parentheses).

Data	Performance measure	Linear GP model	Boosting model	GPBoost model
Non-imputed	RMSE	0.14752 (Benchmark)	0.14044 (−4.80%)	0.14043 (−4.81%)
	CRPS	0.07999 (Benchmark)	0.07824 (−2.19%)	0.07887 (−1.40%)
Non-imputed without spatial variables	RMSE	0.15148 (Benchmark)	0.14641 (−3.35%)	0.14392 (−4.99%)
	CRPS	0.08221 (Benchmark)	0.08109 (−1.36%)	0.07835 (−4.70%)

Note: Without spatial variables means that we omitted all variables related to micro-location, macro-location, regional distinctions, and regional time trends (AQREG_KR).

4.5 | Robustness checks

In the robustness checks described in this section, we perform additional experiments to evaluate the stability and generalizability of our models. First, we check our results based on the non-imputed data by discarding observations with missing values to assess the robustness of our findings to the imputation step. Furthermore, we implement a temporally expanding-window train-test split, where the training window is iteratively expanded by adding an additional quarter and designating the quarter immediately following as test data. This approach allows us to assess model performance over different time periods. Additionally, we conduct a random train-test split, selecting 80% of the entire dataset as the training set and the remaining 20% as the test set. For the GPBoost model (5), we simplify the tuning process by only searching for the optimal number of boosting iterations K , while fixing all other tuning parameters to the values identified during the grid search in the main results. This decision was made to reduce computational complexity. In contrast, for the independent boosting model, all hyperparameters continue to be tuned as in the main analysis. Furthermore, we evaluate robust versions of the models on a significantly smaller dataset comprising listings of German condominiums offered for sale.

4.5.1 | Non-imputed data

To analyze the impact of our missing value imputation procedure, we first check the robustness of our findings using only the non-imputed data. Specifically, we exclude the variables *year of modernization* and *number of floors/floors*, which frequently contain missing values and, as shown in the variable importance analysis in Section 4.3, contribute little to the model. After removing these variables and discarding all remaining observations with missing values, we still retain a dataset with approximately 500,000 observations.

Table 10 reports the RMSE and CRPS of the different models for the non-imputed dataset, both with and without spatial variables. The results closely mirror the main findings presented in Table 4. Furthermore, Figure B3 in Appendix B displays the SHAP dependence plots for *usable space*, *standard*, *condition*, and *year of construction*, based on the non-imputed dataset. These plots closely resemble those shown in Figure 5, indicating that the observed patterns are robust to the imputation procedure.

TABLE 11 Root mean square error (RMSE) and continuous ranked probability score (CRPS) of various models for data with all geocoding and without spatial variables (percentage decrease of RMSE and CRPS in parentheses).

Test quarter	Performance measure	Linear GP model	Boosting model	GPBoost model
Q4 2021	RMSE	0.18021 (Benchmark)	0.16686 (−7.41%)	0.16417 (−8.90%)
	CRPS	0.09933 (Benchmark)	0.09184 (−7.54%)	0.08971 (−9.68%)
Q1 2022	RMSE	0.17173 (Benchmark)	0.16484 (−4.01%)	0.16261 (−5.61%)
	CRPS	0.09421 (Benchmark)	0.09054 (−3.90%)	0.08911 (−5.41%)
Q2 2022	RMSE	0.17809 (Benchmark)	0.16468 (−7.53%)	0.16287 (−9.34%)
	CRPS	0.09780 (Benchmark)	0.09029 (−7.68%)	0.08926 (−8.73%)
Q3 2022	RMSE	0.17156 (Benchmark)	0.16254 (−5.26%)	0.16162 (−5.79%)
	CRPS	0.09374 (Benchmark)	0.08854 (−5.55%)	0.08798 (−6.14%)

Note: Without spatial variables means that we omitted all variables related to micro-location, macro-location, regional distinctions, and regional time trends (AQREG_KR).

Abbreviation: GP, Gaussian process.

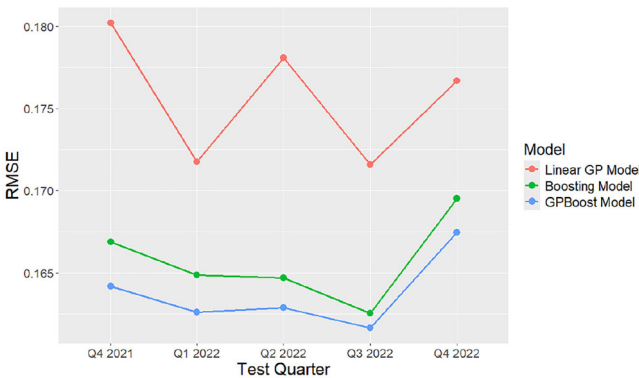


FIGURE 8 Root mean square error (RMSE) for different time periods. The plot presents the RMSEs for linear (Gaussian process) GP model (2), boosting model (4), and GPBoost model (5), tested on various quarters, with training conducted on the periods preceding each test quarter.

4.5.2 | Expanding window train-test split

A temporally expanding window approach allows us to evaluate model performance over different time periods, providing insights into its temporal stability. The results of the moving window analysis are presented in Table 11, where the “Test Quarter” column specifies the quarter on which the model is tested, with the training period comprising all data before that quarter.

In Figure 8, we visualize the results presented in Table 11, including the results for the final time period reported previously. Overall, the RMSE of independent boosting and the GPBoost models decrease over time as the training set size increases, with the exception of the last quarter (Q4 2022). A potential explanation for this could be that the test set for Q4 2022 contains a lower percentage of data points with exact geocoding, resulting in more imputed data compared to the other test quarters. Additionally, the higher RMSE in this quarter may be influenced by changes in the underlying economic conditions, particularly the increase in the interest rate level during this period, which may influence apartment rents.

4.5.3 | Random train-test split

We also repeat the analysis by randomly splitting the data into 80% training and 20% test datasets, mitigating potential bias from temporal or spatial dependencies. The results of this random split,

TABLE 12 Root mean square error (RMSE) and continuous ranked probability score (CRPS) of various models for data with all geocoding and without spatial variables (percentage decrease of RMSE and CRPS in parentheses).

Performance measure	Linear GP model	Boosting model	GPBoost model
RMSE	0.17009 (Benchmark)	0.16055 (−5.61%)	0.15938 (−6.30%)
CRPS	0.09299 (Benchmark)	0.08744 (−5.97%)	0.08668 (−6.79%)

Note: Without spatial variables means that we omitted all variables related to micro-location, macro-location, regional distinctions, and regional time trends (AQREG_KR).
Abbreviation: GP, Gaussian process.

TABLE 13 Root mean square error (RMSE) of various models for data with exact geocoding (percentage decrease of RMSE in parentheses)

Linear GP	Linear GP robust	Boosting	GPBoost	GPBoost robust
0.23891 (Benchmark)	0.24460 (+2.38%)	0.21902 (−8.33%)	0.21701 (−9.17%)	0.22077 (−7.59%)

Abbreviation: GP, Gaussian process.

presented in Table 12, highlight the superior performance of the GPBoost model in terms of RMSE and CRPS, outperforming both the linear GP and the pure boosting models.

The robustness checks, using both time-based moving windows and random splits, confirm the models’ stable performances across different train-test splits. The results align with the main findings, showing the models generalize well even with varied data splits, reinforcing confidence in the robustness and generalizability of our results.

4.6 | Condominium data set

We also analyze a separate dataset comprising listings of German condominiums offered for sale. The data preparation process follows the detailed methodology described in Section 2. For this evaluation, we focus exclusively on listings with exact geocoding and include all available variables, resulting in a smaller training dataset of 79,904 data points—compared to the larger dataset of 1,242,118 training data points used for rents.

Moreover, we extend our comparison to a robust version of the GP models by replacing the Gaussian likelihood with a Student-*t* likelihood (Jylänki et al., 2011), which better captures heavy-tailed residuals and enhances resistance to outliers. Since the current implementation of the GPBoost library <https://github.com/fabsig/GPBoost> does not support this robust approach with the full-scale approximation, we employ the Vecchia approximation (Katzfuss & Guinness, 2021) to efficiently approximate the GP component. This allows us to assess the impact of heavy-tailed modeling assumptions on predictive performance while maintaining computational feasibility. The results, summarized in Table 13, indicate that GPBoost consistently outperforms the other methods, while its robust variant exhibits slightly lower performance compared to the standard GPBoost version.

5 | CONCLUSION

Accurate prediction of rental prices is crucial in the real estate market, and this study highlights the significant improvements that machine learning techniques offer over traditional linear mod-

els. We explore various models, including linear regression, Gaussian processes, tree-boosting, and the hybrid GPBoost model to predict hedonic rents in a large dataset of German residential rental listings.

Our analysis demonstrates that incorporating a GP into linear models significantly enhances the model's ability to capture spatial dependencies. Furthermore, the GPBoost model, which combines the strengths of both Gaussian processes and tree-boosting, consistently outperforms other models in terms of prediction accuracy (RMSE and CRPS). This underscores the importance of accounting for both spatial dependencies and nonlinear patterns. Further, GPBoost maintains its high prediction accuracy even when all spatial covariates are excluded, further emphasizing its robustness across various configurations. These findings are reinforced by multiple robustness checks, confirming the stability and generalizability of the model's superior performance across different train-test data splits and configurations. Additionally, we extend our comparison to robust versions of the models and apply them to a separate dataset of German condominiums offered for sale, where the GPBoost method continues to demonstrate superior performance compared to the other approaches.

Additionally, our analysis using SHAP for model explainability reveals that usable space is the most influential variable in predicting rental prices. Other structure-related factors, such as heating system, condition, and year of construction, also matter, but to a lesser extent. Furthermore, the SHAP dependence plots reveal that the transformations applied in the linear models, such as logarithmic adjustments for usable space and a quadratic term for year of construction, do not fully align with the relationships learned by the boosting algorithms. Consequently, boosting models improve prediction accuracy, particularly in segments where these transformations diverge significantly from the relationships captured by the boosting algorithms. Moreover, boosting models excel at learning complex interactions. For instance, the macro-location variable, annual wage/income tax per taxpayer, exhibits significant interaction effects, especially with usable space, showing a greater impact on smaller apartments.

By incorporating expert knowledge through monotone constraints in tree-boosting models, we enforce prior beliefs about the relationships between predictors and the target variable. We observe that while constrained models slightly underperform in terms of test RMSE and CRPS compared to unconstrained models, the performance difference is small. These constrained models offer advantages in terms of practical applicability and alignment with domain knowledge improving trust and transparency in predictions.

A potential limitation of our analysis, and many other studies, is that the model is trained on asking rents from online listings, which reflect landlords' pricing intentions rather than the amounts ultimately paid by tenants. As Lorenz et al. (2023) note for the German rental market, asking rents are often used due to their superior availability and spatial coverage of listing data, but it carries the risk of overestimating achievable rents because it does not account for post-listing negotiations, rent-free periods, or other concessions. Evidence from Switzerland suggests the magnitude of this bias is context-dependent: Fleury (2018) finds that approximately 82% of dwellings are let at the advertised price, 4% above, and 15% below, with reductions more common in softer markets and higher-priced segments. Together, these findings indicate that while asking rents are a reasonable proxy for actual market conditions in balanced or tight markets, care should be taken when interpreting them in oversupplied markets or for premium segments, where the gap between asking and achieved rents may be substantial. Consequently, SHAP values explain variation in advertised prices and may therefore overstate the economic significance of certain features, particularly in oversupplied markets or premium segments where negotiated discounts are more frequent. Future work could address this by incorporating micro-level rental contract data where available, even if limited to specific sub-markets within Germany, allowing



for targeted validation of model predictions and interpretability results. Such sub-market analyses could help quantify rent gap between asking and actual rents and improve the accuracy of estimated feature effects in monetary terms.

The main focus of this article was on prediction accuracy and not on uncovering causal effects. However, more accurate models can also more accurately estimate causal effects (Varian, 2014), and machine learning methods with high prediction accuracy have helped to develop new theories (De Prado, 2020). In addition, there is a growing literature on extending machine learning models such as random forests (Wager & Athey, 2018), tree-boosting (Bakhitov & Singh, 2022), neural networks (Hartford et al., 2017; Shi et al., 2019), and Bayesian additive regression trees (Hill, 2011) for causal analyses. Future research can analyze how similar applications and extensions of the GPBoost model may be beneficial given its high prediction accuracy.

Looking ahead, there are several exciting directions for future work. Although expanding-window validation addresses temporal robustness to some extent, the modeling approach remains fundamentally cross-sectional and does not capture the evolution of spatial relationships over time. The development of spatial-temporal models, such as incorporating spatio-temporal Gaussian process models (Cressie and Wikle, 2011), could be highly valuable when data over a longer period become available. These models would allow for the investigation of how the relationships between spatial variables and rental prices evolve over time. Analyzing how SHAP dependence and variable importance change as market dynamics shift would provide deeper insights into the underlying trends and factors influencing rental prices. This temporal perspective could further enhance the explainability and robustness of machine learning models in the real estate domain.

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DISCLOSURE

The authors used ChatGPT (OpenAI, San Francisco, CA; version [08/2025]) to assist in improving the language and writing style of this manuscript. All content was reviewed and verified by the authors, who take full responsibility for the final version of the work.

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APPENDIX A: Further Information

A.1 | Transformation of variables

We apply some data-transformation to the listings data from Value AG. These data processing steps prior to fitting a model prove useful, when fitting a linear model. For example, the number of floors is unlikely to have a linear effect on rent, as both the ground floor and the top floor are usually preferred over the first floor. Therefore, forming categories of floor levels might increase

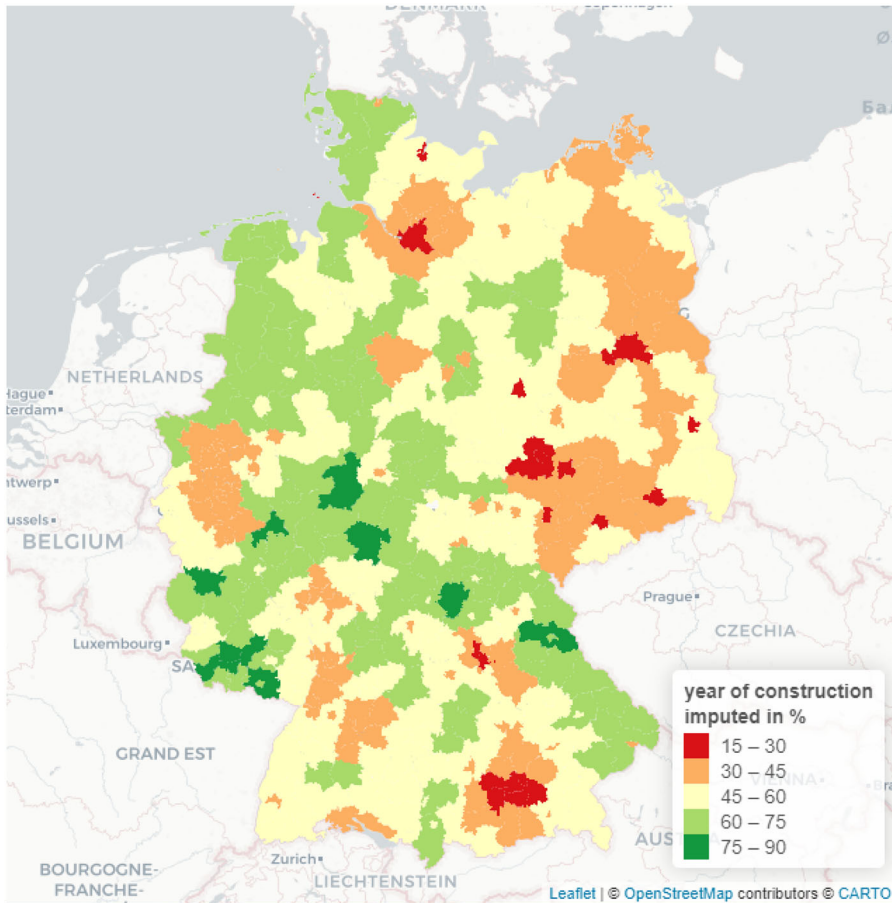


FIGURE A1 Share of imputed year of construction on Kreis-level.

prediction performance in a linear model.

- Location in the building (ARTMIETC): floor categories (floors below 0 and above 6 are in one category, respectively).
- Condition (ZUST): combination of condition in classes from Value AG, year of construction, year of modernization, first occupancy, refurbished, and refurbished required.
- Renovations (RENOV): difference between actual and theoretical building condition in classes using the year of construction to determine the theoretical condition.
- Publication date (TPRJQU): exact publication date saved as combination of year and quarter.

- Regional time trends (AQREG_KR): combination of Kreis number and a dummy indicating the last quarter used in estimation.

A.2 | Imputation of year of construction

- Many regions in western Germany have more than 60% missing values for year of construction.
- The missForest algorithm uses the mean of year of construction as first imputation step. Subsequently, for each variable with missing values, MissForest employs a random forest model on the observed data and predicts the missing values. Figure A2 shows that using the observed

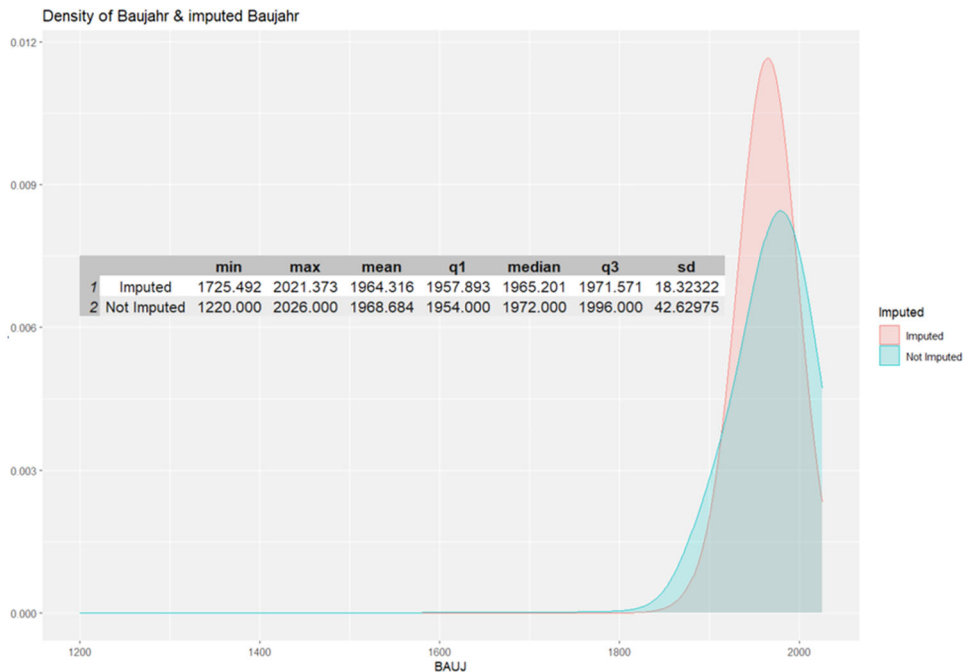


FIGURE A2 Distribution of year of construction.

data, the random forest model was able to generate much variation in year of construction and that the mean and median in the imputed data are lower than in the observed data.

- Variables such as condition, first occupancy, dominant construction period of nearby apartments, flag for old apartment can be valuable to achieve a good imputation.
- We introduced a new dummy variable to flag the imputation of year of construction.
- One might fear that an inaccurate imputation might lead to distorted predictions. We predicted rents for the dataset with all geocoding using the Boosting model (4) once for all observations and once only for observations with available year of construction. Then, we calculated the RMSE for both models on the test data with available year of construction. Using also the observation with imputed year of construction increases the RMSE by 0.4%.

APPENDIX B: Figures and Tables

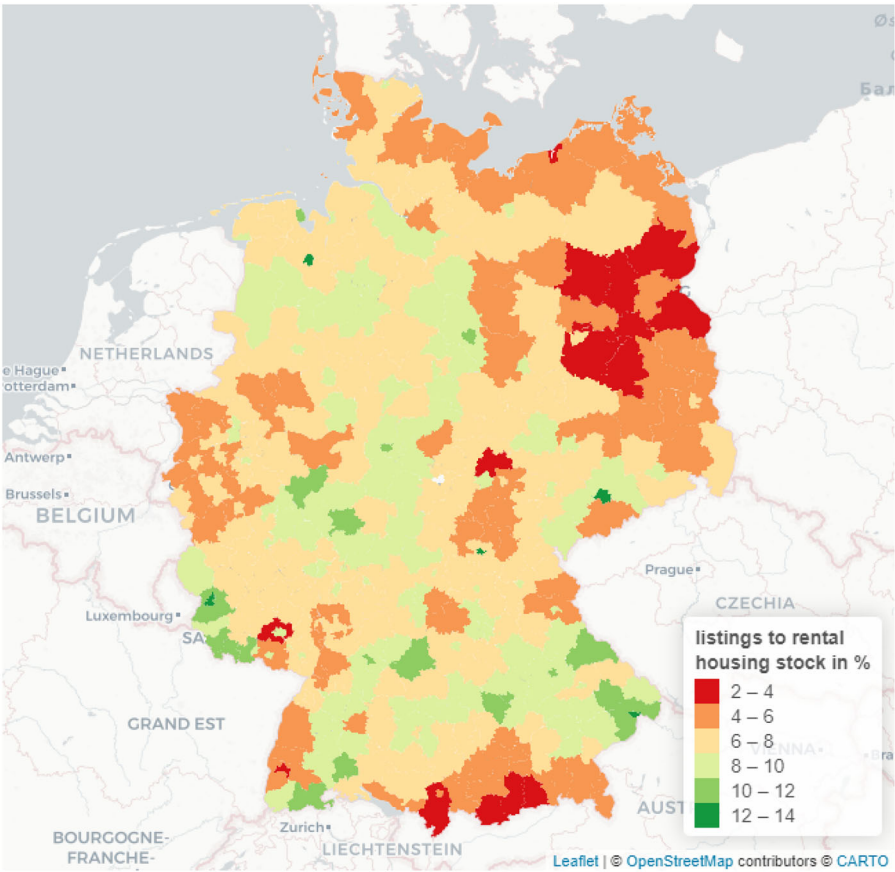


FIGURE B1 Distribution of rental apartment listings relative to rental apartment housing stock on Kreis level.

FIGURE B2 SHAP (SHapley Additive exPlanations) variable importance (exact geocoding). The numbers in parentheses indicate data categories: (1) structural data, (2) macro-location data, (3) micro-location data, (4) regionalization, and (5) other.

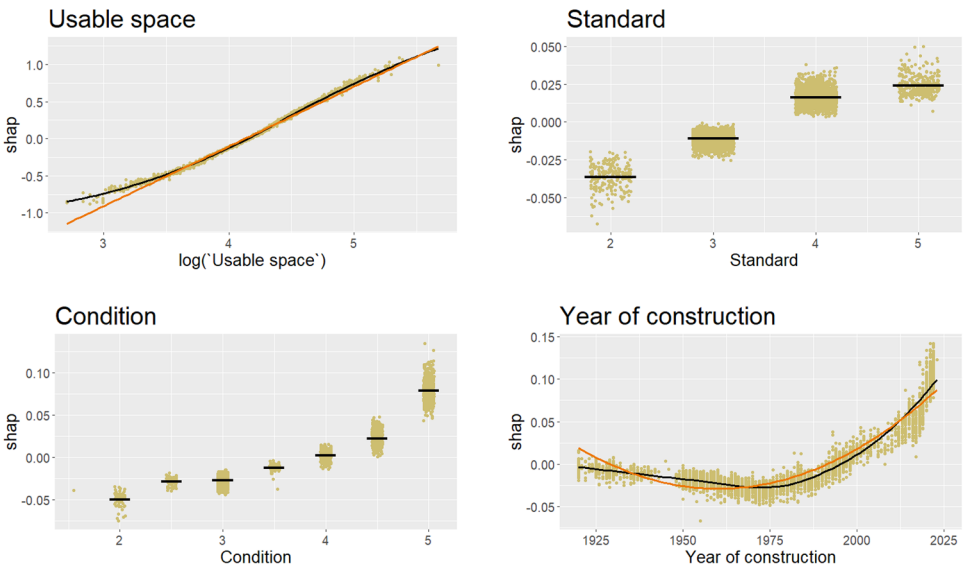
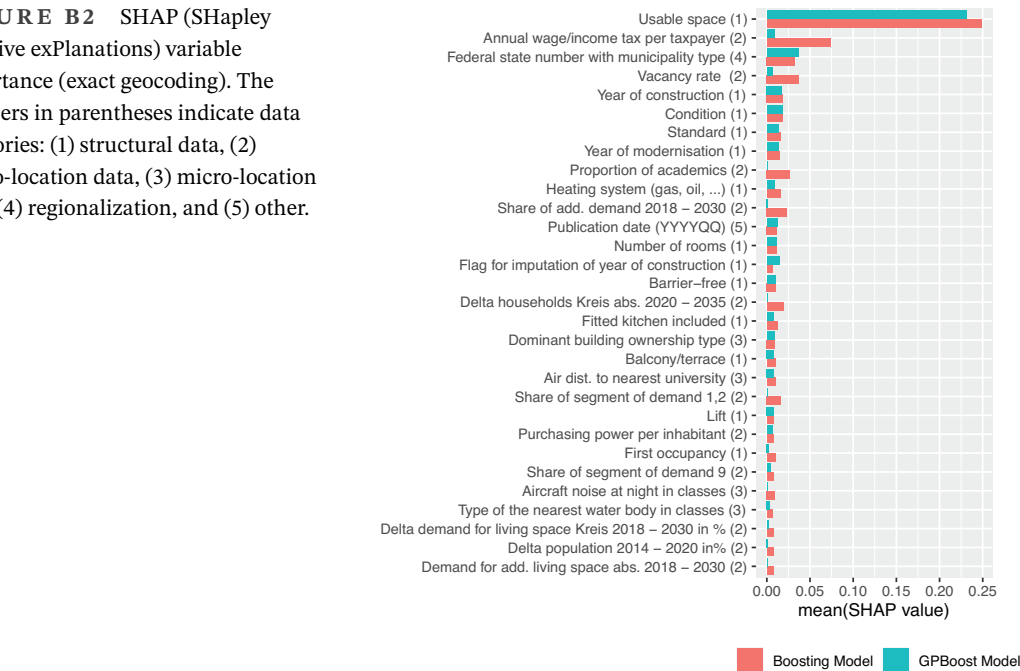


FIGURE B3 SHAP (SHapley Additive exPlanations) dependence plots for important structural data. The plots show SHAP values generated for the GPBoost model (5) trained on the non-imputed dataset and were calculated on a random subsample of 5000 observations to reduce computing time. For “usable space” and “year of construction,” a LOESS (locally estimated scatterplot smoothing) line (in black) highlights the underlying trends. Furthermore, the orange lines depict a linear fit for “usable space” and a quadratic fit for “year of construction.” Additionally, for the categorical variables “standard” and “condition,” the group-wise means (in black) are illustrated.

TABLE B1 Description of variables.

Cat.	Attribute name	Attribute description	Attribute type
Structural data	NEMIET	Current net rent	Numeric
	NUF277	Usable space	Numeric
	ZIZ	Number of rooms	Numeric
	RENOV	Renovations (actual vs. theoretical condition)	Numeric
	BAUJ	Year of construction	Numeric
	BAUJ_IMPUTED	Flag for imputation of year of construction	Factor (2 levels)
	ARTMIETC	Location in the building	Factor (9 levels)
	ZUST	Condition	Numeric
	STAND	Standard	Integer
	etage	Floor	Numeric
	etagen	Number of floors/floors	Numeric
	oeig_neubau_janein	New building	Factor (2 levels)
	zust_projektiert_janein	Projected	Factor (2 levels)
	oeig_altbau_janein	Old building	Factor (2 levels)
	zust_erstbezug_janein	First occupancy	Factor (2 levels)
	zust_saniert_janein	Refurbished	Factor (2 levels)
	zust_sanierungsbedarf_janein	Refurbishment required	Factor (2 levels)
	zust_modernisierungsjahr	Year of modernization	Numeric
	aus_bad_gaestewc_janein	Guest WC	Factor (2 levels)
	aus_barrierefrei_janein	Barrier-free	Factor (2 levels)
	aus_balkon_terrasse_janein	Balcony/terrace	Factor (2 levels)
	oeig_keller_janein	Cellar	Factor (2 levels)
	aus_kueche_einbau_janein	Fitted kitchen included	Factor (2 levels)
	aus_lift_janein	Lift	Factor (2 levels)
	aus_heiz_system	Heating system (central, floor...)	Factor (4 levels)
	aus_heiz_befeuerung	Heating system (gas, oil, ...)	Factor (6 levels)
	aus_parken_janein	Parking facilities	Factor (2 levels)
Macro-location data	KK_EW	Purchasing power per inhabitant	Numeric
	STEU_STPF	Annual wage/income tax per taxpayer	Numeric
	HS_STEU	Property tax rate B	Numeric
	WF_WG	Living space in residential buildings	Numeric
	WF_KREIS	Living space district	Numeric
	D_WHG_EW	Delta apartments per capita 2014–2020	Numeric

(Continues)

TABLE B1 (Continued)

Cat.	Attribute name	Attribute description	Attribute type
	D_BEV	Delta population 2014–2020 abs.	Integer
	D_BEV_PERC	Delta population 2014–2020 in%	Numeric
	ALQ	Unemployment rate	Numeric
	EFH_ANT	Single-family home rate	Numeric
	NASE_12_ANT	Share of segment of demand 1,2	Numeric
	NASE_789_ANT	Share of segment of demand 7,8,9	Numeric
	NASE_369_ANT	Share of segment of demand 3,6,9	Numeric
	NASE_9_ANT	Share of segment of demand 9	Numeric
	OBERZ_MIN	Min. by car to nearest regional centre	Numeric
	BH_MIN	Car min. to IC/EC/ICE stations	Numeric
	ANZ_REGBH	Number of regional railway stations	Integer
	ANZ_FERNBH	Number of long-distance railway stations	Integer
	ANZ_SCHUL	Number of universities	Integer
	ANZ_MESSE	Number of exhibition centres	Integer
	D_HH_KREIS	Delta households Kreis abs. 2020–2035	Integer
	D_HH_KREIS_PERC	Delta households Kreis % 2020–2035	Numeric
	PROG_WOH_KREIS	Demand for add. living space abs. 2018–2030	Integer
	PROG_WOH_ANTEIL_KRE	Share of additional demand 2018–2030	Numeric
	D_PROG_WOH_PERC_KREIS	Delta demand liv. space Kreis 2018–2030 in %	Numeric
	PROG_WOH	Forecasted demand for living space 2030	Numeric
Macro-location data	LEERW_KLASS_KREIS	Vacancy rate	Factor (18 levels)
	GSTADT_DIS	Distance to nearest major city (12 cities)	Numeric
	SvB_AO	Employed persons (SvB) in place of work	Integer
	D_SvB_AO	Delta SvB place of work 2014–2021 abs.	Integer

(Continues)

TABLE B1 (Continued)

Cat.	Attribute name	Attribute description	Attribute type
	D_SvB_AO_PERC	Delta SvB place of work 2014–2021 in %	Numeric
	AKAD_ANT	Proportion of academics	Numeric
	FLUG_DIS	Distance to the airport	Numeric
	PROG_SVB_AO	Forecast SvB place of work 2035	Numeric
	EHKK_EW	Retail purchasing power per inhabitant	Numeric
Micro-location data	RAT_Bild	Education micro-location rating	Numeric
	RAT_Total	Overall micro-location rating	Numeric
	RAT_Wohn	Micro-location rating residential	Numeric
	SUB_Dienstl	Subrating Services	Numeric
	SUB_Freizeit	Subrating Leisure	Numeric
	SUB_Laerm	Subrating Noise pollution	Numeric
	SUB_oev	Subrating Public transport	Numeric
	SUB_Sicht	Subrating View	Numeric
	SUB_Sonne	Subrating Sunlight	Numeric
	SUB_Strasse	Subrating Road access	Numeric
	ERBK_HARZT_PKW_T	Accessibility of general practitioners by car	Numeric
	L_FLUG_DEN_K6	Aircraft noise day/evening/night in classes	Integer
	L_FLUG_N_K6	Aircraft noise at night in classes	Factor (4 levels)
	L_STR_DEN_K6	Road noise day/evening/night in classes	Integer
	L_STR_N_K6	Road noise night in classes	Factor (6 levels)
	L_ZUG_DEN_K6	Railway noise day/evening/night in classes	Integer
	L_ZUG_N_K6	Railway noise night in classes	Factor (6 levels)
	N_EINKZ_DIS	Air dist. to nearest shopping centre	Numeric
	N_KIND_DIS	Air dist. to nearest kindergarten	Numeric
	N_MESSE_DIS	Air dist. to nearest exhibition centre	Numeric

(Continues)

TABLE B1 (Continued)

Cat.	Attribute name	Attribute description	Attribute type
	N_PHARM_DIS	Air dist. to nearest pharmacy	Numeric
	N_REST_DIS	Air dist. to nearest restaurant	Numeric
	N_RET_DIS	Air dist. to nearest grocery store	Numeric
	N_RET_TOT_C_5	Number of grocery shops (500 m radius)	Numeric
	N_SCHUL_DIS	Air dist. to nearest school	Numeric
	N_SCHUL_TOT_C_5	Number of schools (500 m radius)	Numeric
	N_SPITAL_DIS	Air dist. to nearest hospital	Numeric
	N_UNI_DIS	Air dist. to nearest university	Numeric
	OV_HBUS_DIS	Air dist. to third nearest bus stop	Numeric
	OV_HFERN_DIS	Air dist. to nearest ICE station	Numeric
	OV_HFLUG_DIS	Air dist. to nearest internat. airport	Numeric
	OV_HTOT_C_5	Number of public transport stops (500 m radius)	Numeric
	OV_HTRAM_DIS	Air dist. to second closest tram stop	Numeric
	OV_HZUG_DIS	Air dist. to nearest train stop	Numeric
	OV_H_DIS	Air dist. to nearest PT stop	Numeric
	OV_H_DIS2	Air dist. to second nearest PT stop	Numeric
	OV_H_DIS3	Air dist. to third closest PT stop	Numeric
	O_CENT_DIS	Distance to the centre of the location	Numeric
	S_AUT_DIS	Air dist. to nearest motorway entrance	Numeric
	S_TYP1_DIS	Air dist. to nearest motorway	Numeric
	S_TYP2_DIS	Air dist. to nearest inter-regional road	Numeric
	S_TYP_K2	Type of the nearest road (motorway/interurban)	Factor (3 levels)

(Continues)

TABLE B1 (Continued)

Cat.	Attribute name	Attribute description	Attribute type
Micro-location data	T_BCH_DIS	Air dist. to nearest beach	Numeric
	T_EXP_K9	Slope exposure in classes	Numeric
	T_GCLC_DIS	Air dist. to nearest body of water	Numeric
	T_GCLC_TYP_K7	Type of the nearest water body in classes	Factor (8 levels)
	T_GEW_DIS	Distance to the nearest body of water	Numeric
	T_HNG_K13	Slope gradient in classes	Numeric
	T_SGR_DIS	Air dist. to nearest urban green space	Numeric
	T_SUF_DIS	Air dist. to nearest sports / leisure facility	Numeric
	T_WGR_DIS	Air dist. to nearest green space/pasture/meadow	Numeric
	T_WLD_DIS	Air dist. to nearest forest	Numeric
	U_HW_GEF	Flood risk	Integer
	Z_G_DBJ_K10	Dominant construction period of buildings	Numeric
	Z_G_DBJ_R	Prop. of dominant constr. period of buildings	Numeric
	Z_G_DEIG_K8_1	Dominant building ownership type	Factor (9 levels)
	Z_G_DEIG_R_1	Proportion of dominant building ownership type	Numeric
	Z_G_DTYP_K10_1	Dominant building type with size	Factor (11 levels)
	Z_G_DTYP_R_1	Proportion of dominant building type with size	Numeric
	Z_G_TOT_C	Number of buildings per hectare	Numeric
	Z_H_DFTYP_K5_1	Dominant family type according to households	Factor (6 levels)
	Z_H_DLTYP_K7_1	Dominant lifestyle type according to households	Factor (8 levels)
	Z_H_GR1_C	Number of 1-person households	Numeric

(Continues)


TABLE B1 (Continued)

Cat.	Attribute name	Attribute description	Attribute type
	Z_H_GR2_C	Number of 2-person households	Numeric
	Z_H_GR3_C	Number of 3-person households	Numeric
	Z_H_GR4_C	Number of 4-person households	Numeric
	Z_H_GR5_C	Number of 5-person households	Numeric
	Z_H_GR6_C	Number 6 persons Households	Numeric
	Z_H_TOT_C	Number of households per hectare	Numeric
	Z_P_DAGE_K5	Dominant age group per hectare	Numeric
	Z_P_DAGE_R	Proportion of dominant age group per hectare	Numeric
	Z_P_DE_R	Proportion of German citizens per hectare	Numeric
	Z_P_TOT_C	Number of persons per hectare	Numeric
	Z_W_DBJ_K10	Dominant construction period of dwellings	Numeric
	Z_W_DBJ_R	Prop. of dominant construct. period of dwellings	Numeric
	Z_W_DEIG_K5_1	Dominant home ownership type	Factor (6 levels)
	Z_W_DEIG_R_1	Prop. of the dominant residential property type	Numeric
	Z_W_MEDFL_K18_1	Median living space class	Numeric
	Z_W_TOT_C	Number of flats per hectare	Numeric
Regionalization	KreisTyp	BBSR Kreis type	Factor (4 levels)
	GemTyp	Type of municipality (BBSR)	Factor (7 levels)
	GemTypFPRE	Type of municipality (FPRE)	Factor (5 levels)
	BL_GemTyp	Federal state number with municipality type	Factor (86 levels)
Other	x_coord	X-coordinate	Numeric
	y_coord	Y-coordinate	Numeric
	TPRJAQU	Publication date (YYYYQQ)	Numeric
	AQREG_KR	Regional time trends	Factor (393 levels)

TABLE B2 Summary statistics.

Statistic	N	Mean	St. dev.	Min	Max	Imputed by MF in %
NEMIET	1,552,649	8.8	0.5	6.2	11.7	0
x_coord	1,552,649	4,294,149	148,119	4,031,960	4,671,536	0
y_coord	1,552,649	3,085,660	183,964	2,694,082	3,546,805	0
TPRJAQU	1,552,649	20,218	5	20,212	20,223	0
NUF277	1,552,649	70	27	8	600	0
ZIZ	1,552,649	3	1	1	9	0
RENOV	1,552,649	2	1	0	4	44
BAUJ	1,552,649	1967	34	1220	2026	44
ZUST	1,552,649	3.6	0.8	1.5	5	0
STAND	1,552,649	3.4	0.7	2	5	0
RAT_Bild	1,552,649	3.3	0.4	1.6	4.5	36
RAT_Total	1,552,649	2.5	0.6	1.3	4.8	36
RAT_Wohn	1,552,649	2.5	0.6	1.3	4.8	36
SUB_Dienstl	1,552,649	3.4	0.7	1	5	36
SUB_Freizeit	1,552,649	3.7	0.6	1	5	36
SUB_Laerm	1,552,649	3.2	0.6	1.7	4.9	36
SUB_oev	1,552,649	4.2	0.5	1	5	36
SUB_Sicht	1,552,649	2.9	0.4	2	5	36
SUB_Sonne	1,552,649	3.4	0.4	2	5	36
SUB_Strasse	1,552,649	3.8	0.5	1.7	5	36
ERBK_HARZT_PKW_T	1,552,649	2	1	0	60	36
L_FLUG_DEN_K6	1,552,649	0	5	0	70	0
L_STR_DEN_K6	1,552,649	13	25	0	75	0
L_ZUG_DEN_K6	1,552,649	6	18	0	75	0
N_EINKZ_DIS	1,552,649	8633	9123	0	69,263	36
N_KIND_DIS	1,552,649	1636	1776	0	23,988	36
N_MESSE_DIS	1,552,649	38,228	30,520	22	241,066	36
N_PHARM_DIS	1,552,649	864	1023	0	18,850	36
N_REST_DIS	1,552,649	533	559	0	14,123	36
N_RET_DIS	1,552,649	646	778	16	15,168	36
N_RET_TOT_C_5	1,552,649	2	2	0	22	36
N_SCHUL_DIS	1,552,649	776	809	0	16,206	36
N_SCHUL_TOT_C_5	1,552,649	1	1	0	25	36
N_SPITAL_DIS	1,552,649	2815	2516	2	30,155	36
N_UNI_DIS	1,552,649	11,216	11,055	2	90,585	36
OV_HBUS_DIS	1,552,649	209	203	0	9375	36
OV_HFERN_DIS	1,552,649	10,986	12,018	5	92,090	36
OV_HFLUG_DIS	1,552,649	44,082	29,676	175	249,997	36
OV_HTOT_C_5	1,552,649	9	6	0	62	36

(Continues)



TABLE B2 (Continued)

Statistic	N	Mean	St. dev.	Min	Max	Imputed by MF in %
OV_HTRAM_DIS	1,552,649	24,224	28,668	0	209,560	36
OV_HZUG_DIS	1,552,649	1792	1785	2	24,819	36
OV_H_DIS	1,552,649	192	175	0	6486	36
OV_H_DIS2	1,552,649	250	228	3	8276	36
OV_H_DIS3	1,552,649	351	300	5	8759	36
O_CENT_DIS	1,552,649	19,058	19,966	1	142,129	36
S_AUT_DIS	1,552,649	5834	5589	15	68,808	36
S_TYP1_DIS	1,552,649	1076	1098	0	36,805	36
S_TYP2_DIS	1,552,649	1238	1,574	0	19,199	36
T_BCH_DIS	1,552,649	53,394	26,378	0	150,610	36
T_EXP_K9	1,552,649	2	2	0	8	36
T_GCLC_DIS	1,552,649	2914	2414	0	24,646	36
T_GEW_DIS	1,552,649	708	492	0	8764	36
T_HNG_K13	1,552,649	2	1	1	8	36
T_SGR_DIS	1,552,649	2341	3115	0	42,867	36
T_SUF_DIS	1,552,649	871	823	0	12,189	36
T_WGR_DIS	1,552,649	749	717	0	7024	36
T_WLD_DIS	1,552,649	870	680	0	12,537	36
U_HW_GEF	1,552,649	0.1	0.4	0	3	0
Z_G_DBJ_K10	1,552,649	2.9	1.2	1	10	47
Z_G_DBJ_R	1,552,649	0.9	0.2	0.1	1	47
Z_G_DEIG_R_1	1,552,649	0.8	0.2	0.2	1	38
Z_G_DTYP_R_1	1,552,649	0.6	0.2	0.1	1	39
Z_G_TOT_C	1,552,649	7	6	0	141	36
Z_H_GR1_C	1,552,649	20	26	0	499	44
Z_H_GR2_C	1,552,649	11	11	0	163	44
Z_H_GR3_C	1,552,649	4	4	0	64	44
Z_H_GR4_C	1,552,649	2	3	0	56	44
Z_H_GR5_C	1,552,649	1	1	0	37	44
Z_H_GR6_C	1,552,649	0	1	0	31	44
Z_H_TOT_C	1,552,649	34	42	0	572	36
Z_P_DAGE_K5	1,552,649	3	1	1	5	42
Z_P_DAGE_R	1,552,649	0.4	0.2	0.2	1	42
Z_P_DE_R	1,552,649	0.8	0.2	0	1	42
Z_P_TOT_C	1,552,649	63	69	0	1142	36
Z_W_DBJ_K10	1,552,649	3	1	1	10	45
Z_W_DBJ_R	1,552,649	0.8	0.2	0.1	1	45
Z_W_DEIG_R_1	1,552,649	0.8	0.1	0.3	1	38
Z_W_MEDFL_K18_1	1,552,649	6	2	1	17	38

(Continues)

TABLE B2 (Continued)

Statistic	N	Mean	St. dev.	Min	Max	Imputed by MF in %
Z_W_TOT_C	1,552,649	36.7	45.3	0	605	36
KK_EW	1,552,649	24,661	2,992	11,032	63,934	0
STEU_STPF	1,552,649	7228	2402	0	77,555	0
HS_STEU	1,552,649	502	131	80	1050	0
WF_WG	1,552,649	9814	15,225	3	72,917	0
WF_KREIS	1,552,649	19,221	23,884	1630	142,716	0
D_WHG_EW	1,552,649	0	0	−0.4	0.6	0
D_BEV	1,552,649	8964	18,378	−4071	89,687	0
D_BEV_PERC	1,552,649	0	0	−0.4	0.5	0
ALQ	1,552,649	0.1	0	0	0.4	0
EFH_ANT	1,552,649	0.3	0.1	0	0.9	0
NASE_12_ANT	1,552,649	0.2	0	0.1	0.6	0
NASE_789_ANT	1,552,649	0.3	0.1	0.1	0.8	0
NASE_369_ANT	1,552,649	0.4	0.1	0	0.6	0
NASE_9_ANT	1,552,649	0.1	0	0	0.3	0
OBERZ_MIN	1,552,649	13.4	16	0	187.3	0
BH_MIN	1,552,649	13.8	16.1	0	142.1	0
ANZ_REGBH	1,552,649	13.3	26.3	0	154	0
ANZ_FERNBH	1,552,649	1.1	2	0	11	0
ANZ_SCHUL	1,552,649	3.1	6.6	0	38	0
ANZ_MESSE	1,552,649	0.2	0.4	0	1	0
D_HH_KREIS	1,552,649	13,219	32,903	−22,774	177,902	0
D_HH_KREIS_PERC	1,552,649	0	0.1	−0.2	0.2	0
PROG_WOH_KREIS	1,552,649	972	1828	−862	10,561	0
PROG_WOH_ANTEIL_KREIS	1,552,649	0	0	−0.1	0.1	0
D_PROG_WOH_PERC_KREIS	1,552,649	0	0	−0.1	0.1	0
PROG_WOH	1,552,649	10,097	15,909	3	76,081	0
GSTADT_DIS	1,552,649	53	42	0	240	0
SvB_AO	1,552,649	166,449	315,273	0	1,582,539	0
D_SvB_AO	1,552,649	22,160	53,661	−2890	313,114	0
D_SvB_AO_PERC	1,552,649	0.1	0.1	−1	8.9	0
AKAD_ANT	1,552,649	0.2	0.1	0.1	0.4	0
FLUG_DIS	1,552,649	38	23	1	218	0
PROG_SVB_AO	1,552,649	161,053	315,957	0	1,596,853	0
EHKK_EW	1,552,649	6467	572	2767	15,391	0
etage	1,552,649	2	2	−1	49	9
etagen	1,552,649	4	1	1	49	62
zust_modernisierungsjahr	1,552,649	2014	5	1938	2024	81

TABLE B3 Monotone expert constraints, where + denotes a monotone increasing constraint and – denotes a monotone decreasing constraint.

Cat.	Attribute name	Attribute description	Constraints Version 1	Constraints Version 2
Struct. data	NUF277	Usable space	+	+
	RENOV	Renovations (actual vs. theoretical condition)	+	+
	ZUST	Condition	+	+
	STAND	Standard	+	+
	zust_modernisierungsjahr	Year of modernization		+
Micro-location data	RAT_Wohn	Micro-location rating residential	+	+
	SUB_Dienstl	Subrating services		+
	SUB_Freizeit	Subrating leisure		+
	SUB_Laerm	Subrating noise pollution		+
	SUB_oev	Subrating public transport		+
	SUB_Sicht	Subrating view		+
	SUB_Sonne	Subrating sunlight		+
	SUB_Strasse	Subrating road access		+
	L_FLUG_DEN_K6	Aircraft noise day/evening/night in classes		–
	L_STR_DEN_K6	Road noise day/evening/night in classes		–
	L_ZUG_DEN_K6	Railway noise day/evening/night in classes		–
	U_HW_GEF	Flood risk		–

TABLE B4 Transformations of variables for linear models.

Cat.	Attribute name	Attribute description	Transformation
Struct. data	NEMIET	Current net rent	logarithmic
	NUF277	Usable space	logarithmic
	BAUJ	Year of construction	quadratic
Macro-location data	KK_EW	Purchasing power per inhabitant	logarithmic
	STEU_STPF	Annual wage/income tax per taxpayer	logarithmic
	WF_WG	Living space in residential buildings	logarithmic
	WF_KREIS	Living space district	logarithmic
	EFH_ANT	Single-family home rate	arcsine square root
	NASE_12_ANT	Share of segment of demand 1,2	arcsine square root
	NASE_789_ANT	Share of segment of demand 7,8,9	arcsine square root

(Continues)

TABLE B4 (Continued)

Cat.	Attribute name	Attribute description	Transformation
	NASE_369_ANT	Share of segment of demand 3,6,9	arcsine square root
	NASE_9_ANT	Share of segment of demand 9	arcsine square root
	OBERZ_MIN	Min. by car to nearest regional centre	square root
	BH_MIN	Car min. to IC/EC/ICE stations	square root
	ANZ_REGBH	Number of regional railway stations	square root
	ANZ_FERNBH	Number of long-distance railway stations	square root
	ANZ_SCHUL	Number of universities	square root
	ANZ_MESSE	Number of exhibition centres	square root
	PROG_WOH_KREIS	Demand for additional living space abs. 2018 - 2030	log
	GSTADT_DIS	Distance to nearest major city (12 cities)	square root
	SvB_AO	Employed persons (SvB) in place of work	square root
	AKAD_ANT	Proportion of academics	arcsine square root
	FLUG_DIS	Distance to the airport	logarithmic
	PROG_SVB_AO	Forecast SvB place of work 2035	square root
	EKKK_EW	Retail purchasing power per inhabitant	logarithmic
Micro-location data	ERBK_HARZT_PKW_T	Accessibility of general practitioners by car	square root
	N_EINKZ_DIS	Air dist. to nearest shopping centre	logarithmic
	N_KIND_DIS	Air dist. to nearest kindergarten	logarithmic
	N_MESSE_DIS	Air dist. to nearest exhibition centre	logarithmic
	N_PHARM_DIS	Air dist. to nearest pharmacy	logarithmic
	N_REST_DIS	Air dist. to nearest restaurant	logarithmic
	N_RET_DIS	Air dist. to nearest grocery store	logarithmic

(Continues)

TABLE B4 (Continued)

Cat.	Attribute name	Attribute description	Transformation
Micro-location data	N_RET_TOT_C_5	Number of grocery shops (500 m radius)	square root
	N_SCHUL_DIS	Air dist. to nearest school	logarithmic
	N_SCHUL_TOT_C_5	Number of schools (500 m radius)	square root
	N_SPITAL_DIS	Air dist. to nearest hospital	logarithmic
	N_UNI_DIS	Air dist. to nearest university	logarithmic
	OV_HBUS_DIS	Air dist. to third nearest bus stop	logarithmic
	OV_HFERN_DIS	Air dist. to nearest ICE station	logarithmic
	OV_HFLUG_DIS	Air dist. to nearest internat. airport	logarithmic
	OV_HTOT_C_5	Number of public transport stops (500 m radius)	square root
	OV_HTRAM_DIS	Air dist. to second closest tram stop	logarithmic
	OV_HZUG_DIS	Air dist. to nearest train stop	logarithmic
	OV_H_DIS	Air dist. to nearest PT stop	logarithmic
	OV_H_DIS2	Air dist. to second nearest PT stop	logarithmic
	OV_H_DIS3	Air dist. to third closest PT stop	logarithmic
	O_CENT_DIS	Distance to the centre of the location	logarithmic
	S_AUT_DIS	Air dist. to nearest motorway entrance	logarithmic
	S_TYP1_DIS	Air dist. to nearest motorway	logarithmic
	S_TYP2_DIS	Air dist. to nearest inter-regional road	logarithmic
T_BCH_DIS	Air dist. to nearest beach	logarithmic	
T_GCLC_DIS	Air dist. to nearest body of water	logarithmic	
T_GEW_DIS	Distance to the nearest body of water	logarithmic	
T_SGR_DIS	Air dist. to nearest urban green space	logarithmic	

(Continues)

TABLE B4 (Continued)

Cat.	Attribute name	Attribute description	Transformation
	T_SUF_DIS	Air dist. to nearest sports / leisure facility	logarithmic
	T_WGR_DIS	Air dist. to nearest green space/pasture/meadow	logarithmic
	T_WLD_DIS	Air dist. to nearest forest	logarithmic
	Z_G_DBJ_R	Prop. of dominant constr. period of buildings	arcsine square root
	Z_G_DEIG_R_1	Proportion of dominant building ownership type	arcsine square root
	Z_G_DTYP_R_1	Proportion of dominant building type with size	arcsine square root
	Z_G_TOT_C	Number of buildings per hectare	square root
	Z_H_GR1_C	Number of 1-person households	square root
	Z_H_GR2_C	Number of 2-person households	square root
	Z_H_GR3_C	Number of 3-person households	square root
	Z_H_GR4_C	Number of 4-person households	square root
	Z_H_GR5_C	Number of 5-person households	square root
	Z_H_GR6_C	Number 6 persons Households	square root
	Z_H_TOT_C	Number of households per hectare	square root
	Z_P_DAGE_R	Proportion of dominant age group per hectare	arcsine square root
	Z_P_DE_R	Proportion of German citizens per hectare	arcsine square root
	Z_P_TOT_C	Number of persons per hectare	square root
	Z_W_DBJ_R	Prop. of dominant construct. period of dwellings	arcsine square root
	Z_W_DEIG_R_1	Prop. of the dominant residential property type	arcsine square root
	Z_W_MEDFL_K18_1	Median living space class	logarithmic
	Z_W_TOT_C	Number of flats per hectare	square root



TABLE B5 Gaussian process covariance parameters of various models and data versions.

Covariance parameter	All Geocoding		All Geocoding without spatial variables	
	Linear GP model	GPBoost model	Linear GP model	GPBoost model
Nugget effect σ^2	0.0244	0.0168	0.0252	0.0213
Marginal variance σ_1^2	0.0115	0.00818	0.0418	0.0106
Length-scale ρ	3046	21,159	10,433	20,442

Note: Without spatial variables means that we omitted all variables related to micro-location, macro-location, regional distinctions, and regional time trends (AQREG_KR).

Abbreviation: GP, Gaussian process.

TABLE B6 Test results for comparing the different models. The tests are done using a regression model with a random effect at the observation level.

Data	Summary	(Intercept)	Linear model	Linear FE model
Exact geocoding	Estimate	0.02505	0.00491	0.00285
	<i>t</i> value	138.25	56.28	32.62
	<i>p</i> value	0.00***	0.00***	1.06e-232***
Exact geocoding without spatial variables	Estimate	0.02614	0.05279	0.00257
	<i>t</i> value	92.83	168.98	8.21
	<i>p</i> value	0.00***	0.00***	2.22e-16***
All geocoding	Estimate	0.03029	0.00359	0.001179
	<i>t</i> value	219.76	73.90	24.28
	<i>p</i> value	0.00***	0.00***	4.01e-130***
All geocoding without spatial variables	Estimate	0.03121	0.04908	0.00080
	<i>t</i> value	155.45	240.87	3.91
	<i>p</i> value	0.00***	0.00***	9.41e-05***

Note: Without spatial variables means that we omitted all variables related to micro-location, macro-location, regional distinctions, and regional time trends (AQREG_KR). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE B7 Comparison of model performance (all geocoding).

Measure	Linear model	Linear FE model	Linear GP model	Boosting model	GPBoost model
MAPE	14.22%	13.55%	13.16%	12.66%	12.85%
MAPE applied to avg. monthly rent in EUR	86.39	87.88	85.35	82.11	83.34
MdAPE	11.10%	10.53%	10.1%	9.69%	9.82%
Within 20%	76.2%	78.2%	79.4%	80.9%	80.5%
R^2	0.851	0.862	0.867	0.875	0.875

Note: All measures were calculated on the data with all geocoding with spatial variables.

Abbreviation: GP, Gaussian process; MAPE, mean absolute percentage error; MdAPE, median absolute percentage error.

TABLE B8 Test results for comparing boosting models with and without expert constraints. The tests are done using a regression model with a random effect at the observation level.

Data	Summary	(Intercept)	Boosting model with constraint version 1	Boosting model with constraint version 2
Exact Geocoding	Estimate	0.02225	0.00043	0.00080
	<i>t</i> value	136.85	6.85	12.69
	<i>p</i> value	0.00***	7.66e-12***	7.06e-37***

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.